

**Generative Discovery and First-Principles Validation
of
Economical High-Entropy Carbide Electrocatalysts
for CO₂ Electroreduction (CO₂RR)**

— from CO-generating active-site design to multi-carbon products

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Abstract

Electrochemical CO₂ reduction (CO₂RR) is a key route for closing the anthropogenic carbon cycle, converting CO₂ with renewable electricity into useful carbon compounds—from carbon monoxide (CO) to multicarbon (C₂₊) fuels such as ethanol—yet the existing highly selective catalysts (Au, Ag, etc.) are scarce and expensive. Here CO is the *gateway intermediate* en route to more deeply reduced products, so the suppression of the competing hydrogen evolution reaction (HER) alongside efficient CO formation is a shared necessary condition for all CO₂RR products. This thesis develops and applies an end-to-end *generative* discovery pipeline that searches directly within the chemical space of high-entropy transition-metal carbides (high-entropy carbide, HEC) for *affordable* CO₂RR carbide catalysts. A property-conditioned diffusion model (a fine-tuned MATTERGEN) is guided toward compositions that are stable, metallic, CO-releasing, and free of platinum-group metals, and the generated candidate pool is passed through a funnel of composition-based surrogate descriptors, surface free-energy screening with a machine-learning interatomic potential (UMA), and density-functional theory (DFT) validation in QUANTUM ESPRESSO. By driving a single pipeline with *two constraint sets*, we perform a controlled comparison of how the four design axioms that an excellent electrocatalyst must satisfy (activity, selectivity, manufacturability, and affordability) interact at the generation stage. *Campaign A* (the activity-first baseline) converges on six platinum-group-free HECs, but these are unmanufacturable because rare-earth elements dissolve during the essential leaching step of metathermic-reduction–*acid-leaching* synthesis—demonstrating that unconstrained activity search *heads toward unmanufacturable chemistry*. *Campaign B* imposes all four axioms: it restricts the search space to acid-leaching-stable group-4/5/6 transition metals (excluding rare earths and platinum-group metals, and also excluding the known CO₂RR metals Cu/Ag/Au/Zn to ensure that the discovery is genuinely contributed by the generative model), conditions on surface labels computed with UMA, and treats HER competition explicitly—because strong hydrogen adsorption is *poisoning* that blocks the CO₂ site—by applying a *dual active-site* criterion that simultaneously requires CO formation ($|\Delta G_{\text{CO}^*}| < 0.3 \text{ eV}$) and *H repulsion* ($\Delta G_{\text{H}^*} > 0$) on the same facet. Rare earths are excluded for manufacturability reasons, and active-site analysis shows that they were *unnecessary in the first place* (the dual behavior arises from the valence-electron concentration (VEC) effect of the transition metals). As a result, three Ti-based three-metal rocksalt carbides (hec3-MoTiZr/CrTiV/CrTiZr, (100)) pass the DFT dual objective, and we confirm that H poisoning is the dominant failure mode and that a low VEC is the key tuning lever. Full ionic relaxation of the representative candidate hec3-MoTiZr confirms its selectivity ($\Delta G_{\text{H}^*} = +0.57$, $\Delta G_{\text{CO}^*} = +0.18 \text{ eV}$) while correcting the limiting potential to $U_L = -0.40 \text{ V}$, and re-aiming the generator toward low VEC caused the generative model itself to produce H-repelling candidates. To complement this, we design and present a plan for experimental validation—materials synthesis, electrochemical CO₂RR testing, online gas analysis of CO/CO₂, ¹H NMR of liquid-phase products, and (in-situ) FT-IR of adsorbed intermediates. In preliminary electrolysis, *ethanol formation was observed by ¹H NMR* (quantification of Faradaic efficiency, operating conditions, etc. is **[quantitative analysis – to be updated upon experimental completion]**), which suggests that, beyond the CO formation the design targeted, the *multi-site tandem* of a high-entropy surface (a CO-forming site plus a C–C coupling site) can lead to multicarbon products (first-principles calculation of the C–C coupling pathway is future

work). By showing that affordability, synthesizability, and HER selectivity can be internalized together *at the generation stage*, this work shifts catalyst discovery from the screening of known materials to the *design* of affordable new materials.

Keywords: CO₂ reduction (CO₂RR); high-entropy carbide; generative model; machine-learning interatomic potential; density-functional theory; non-noble-metal electrocatalyst; HER selectivity; multicarbon (ethanol) product.

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Chapter 1

Introduction

1.1 Motivation: Affordable CO₂-Reduction Catalysts

Electrochemical CO₂ reduction (CO₂RR) driven by renewable energy offers a route to convert greenhouse-gas waste into chemical feedstocks. Among the accessible products, carbon monoxide is one of the most technologically valuable: it is the two-electron product with the highest demonstrated selectivity and the starting point for Fischer–Tropsch and carbonylation chemistry. Moreover, CO is also the *gateway intermediate* toward high-value multicarbon (C₂₊) fuels such as ethanol via electrochemical C–C coupling—that is, efficient CO formation and the suppression of competing HER are a shared prerequisite not only for CO itself but also for the C₂₊ products beyond it, and the catalyst this study targets is an affordable surface that satisfies these two necessary conditions. The CO selectivity of a metal electrode is determined by how strongly it binds the key intermediates *COOH and *CO relative to adsorbed hydrogen *H. According to Hori’s classic survey of metal electrodes, Au, Ag, and Zn are the representative CO-producing metals, Cu uniquely proceeds all the way to hydrocarbons, and most other metals favor the competing hydrogen evolution reaction (HER) [9]. The mechanistic rationale for this behavior on copper and the central role of the *COOH and *CO descriptors were comprehensively synthesized by Nitopi et al. [8].

The problem is economic rather than chemical. The most CO-selective metals (Au, Ag) are scarce and expensive, which is hard to reconcile with the terawatt scale at which CO₂ conversion must operate if it is to matter for the climate. There is therefore an urgent need for catalysts that reproduce the favorable adsorption-energy landscape of the coinage metals using earth-abundant elements.

The copper paradox: cheap but not enough. In the context of multicarbon (C₂₊) products, price is not the immediate bottleneck—copper is cheap and yet is essentially the only metal that carries out C–C coupling beyond CO to ethylene and ethanol [8]. It is then natural to ask, “why design new materials instead of just using copper?” The answer is that copper has intrinsic limitations that block commercialization: (i) *low product selectivity*—ethylene, ethanol, *n*-propanol, methane, formic acid, CO, and H₂ are produced together at a single potential, so the Faradaic efficiency toward any single product is low and separation costs are large; (ii) *structural instability during operation*—the metal surface undergoes reconstruction, coarsening, and oxidation–reduction cycling, so its active sites and selectivity change over time; (iii) still-large overpotentials and residual HER competition [8]. Moreover, being a *single metal*, copper has an essentially fixed adsorption landscape at its active sites, leaving little room to tune selectivity through composition. In short, the bottleneck depends on the reaction: for CO formation it is *price* (Au/Ag), and for C₂₊ formation it is *selectivity and stability* (Cu); in both cases the fundamental constraint is the *fixed adsorption landscape of a single metal*. This is why, instead of a single metal, this study chooses the high-entropy carbide platform (next section), whose

adsorption landscape can be tuned continuously through composition and which is intrinsically stable.

1.2 Why High-Entropy Carbides

This thesis combines two materials-design ideas. First, transition-metal carbides (TMCs) have long been known to exhibit “platinum-like” surface chemistry. The pioneering observation by Levy and Boudart that tungsten carbide mimics platinum in several catalytic reactions [12] sparked decades of research viewing TMCs as affordable substitutes for noble metals; in particular, molybdenum carbide is an established selective catalyst for reducing CO₂ to CO [13]. Carbon donates electron density into the metal *d* band and reconstructs the surface density of states, tuning the adsorption strength into a useful range.

Second, *high-entropy* materials—nearly equimolar solid solutions of four or more elements—provide a nearly continuous distribution of local adsorption environments. Batchelor et al. argued that this continuity is exactly what electrocatalysis needs, because a small number of optimal binding sites can dominate the measured activity [10]. High-entropy alloys were subsequently realized experimentally as CO₂/CO reduction catalysts [11]. High-entropy carbides are themselves synthesizable single-phase materials, as demonstrated by the discovery of refractory five-metal carbides using an entropy descriptor [14]. Third, as *ceramics*, carbides are chemically and thermally robust, providing a stable framework that can in principle avoid the in-operation reconstruction and degradation problems of copper noted above.

These three axes each address one of copper’s three limitations: the “platinum-like” electronic structure provides *activity*, the high-entropy site diversity provides *compositional tuning* of the adsorption landscape (selectivity, and room for multi-site tandem design, Section 6.6), and the ceramic framework provides *operational stability*. Restricting the palette to affordable metals adds *affordability* on top—this combination defines the search space of this study. That said, the calculations in this study only validate CO formation and HER suppression; they do not prove a C₂₊ selectivity *advantage* over copper; that remains a platform hypothesis to be judged by experiment and future coupling calculations.

1.3 Why a Generative Pipeline

Conventional computational catalyst discovery screens *already-existing* databases. The combinatorial space of multi-metal carbides is too vast to enumerate, and within it the affordable subset is sparse. This study instead adopts a *generative* strategy. A diffusion model that has learned the distribution of stable inorganic crystals [1] is fine-tuned so that sampling is *guided* toward target property values, including an explicit *affordability* constraint that penalizes platinum-group and heavy-rare-earth content. Affordability thus becomes a generation objective rather than a post-hoc filter. Moreover, the conditioning is not limited to affordability: by jointly targeting activity, selectivity (H repulsion), and manufacturability (the four axioms of Chapter 4), we perform multi-objective composition optimization *that is impossible for a single metal* across the vast space of multi-metal carbides—precisely the degree of freedom that copper’s fixed adsorption landscape cannot provide. Because the model proposes new structures, the expensive part of discovery (DFT) is used only on the few promising candidates reached through a funnel of increasingly accurate but increasingly costly evaluators.

1.4 Objectives and Structure

This thesis investigates how the four design axioms that an excellent CO₂RR carbide electrocatalyst must satisfy—**activity** (A1: CO release, $\Delta G_{\text{CO}^*} \approx 0$), **selectivity** (A2: HER suppression,

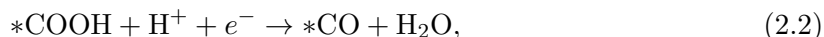
H repulsion), **manufacturability** (A3: acid-leaching survival), and **affordability** (A4: noble-metal-free)—can be jointly satisfied at the generation stage. The specific objectives are as follows. (i) Build a generative pipeline that takes these axioms as conditioning properties. (ii) Perform a controlled comparison by driving a single pipeline with *two constraint sets*—*Campaign A* (the activity-first baseline) reveals where unconstrained activity search converges, and *Campaign B* (fully constrained, CO/H dual active-site) yields candidates that satisfy all four axioms. (iii) Validate the top candidates with first-principles DFT for stability, metallic conductivity, and surface free energy, and identify the physical lever (valence-electron concentration, VEC) that governs the active-site behavior. (iv) Design an experimental campaign to confirm the computational predictions. Chapter 2 covers the theoretical background, Chapter 3 the computational methodology, Chapter 4 the computational results, Chapter 5 the planned experimental validation, Chapter 6 the discussion, and Chapter 7 the conclusions.

Chapter 2

Theoretical Background

2.1 CO₂-to-CO Thermodynamics and the Computational Hydrogen Electrode

The two-electron pathway to carbon monoxide proceeds through a single carboxyl intermediate.



The reaction free energies of the electrochemical steps are evaluated with the computational hydrogen electrode (CHE) of Nørskov et al. [7], in which the chemical potential of the proton-electron pair is referenced to gas-phase H₂ at 0 V versus the reversible hydrogen electrode.

$$\mu(\text{H}^+) + \mu(e^-) = \frac{1}{2} \mu(\text{H}_2) - eU. \quad (2.4)$$

The adsorption free energies used throughout this work are as follows.

$$\Delta G_{\text{CO}*} = E(*\text{CO}) - E(*) - E(\text{CO}_{(\text{g})}) + \Delta_{\text{CO}}, \quad (2.5)$$

$$\Delta G_{\text{COOH}*} = E(*\text{COOH}) - E(*) - E(\text{CO}_{2(\text{g})}) - \frac{1}{2} E(\text{H}_{2(\text{g})}) + \Delta_{\text{COOH}}, \quad (2.6)$$

$$\Delta G_{\text{H}*} = E(*\text{H}) - E(*) - \frac{1}{2} E(\text{H}_{2(\text{g})}) + \Delta_{\text{H}}, \quad (2.7)$$

where the Δ terms are the standard zero-point-vibrational and entropic corrections. A good carbon-monoxide-producing site requires $\Delta G_{\text{CO}*}$ to be close to zero (so that carbon monoxide is released as product rather than poisoning the surface), $\Delta G_{\text{COOH}*}$ to be low enough to activate carbon dioxide, and—crucially for selectivity— $\Delta G_{\text{COOH}*} < \Delta G_{\text{H}*}$ (so that CO₂RR outcompetes HER).

2.2 Descriptors, the Sabatier Principle, and Carbides

Because adsorption energies on the relevant surfaces scale with one another, a single descriptor (here the $\Delta G_{\text{CO}*}$ of the optimal site) captures most of the activity trend, with the optimum lying at the Sabatier maximum of intermediate binding strength. In carbides, the carbon p states hybridize with the metal d states and shift the d -band; this is the microscopic origin of the “platinum-like” behavior of WC [12] and of the carbon-monoxide selectivity of Mo₂C [13]. Because a metallic density of states at the Fermi level is a prerequisite for an electrocatalyst, verifying metallicity with DFT is an indispensable validation step.

2.3 Generative Diffusion Models for Crystals

MATTERGEN [1] is a diffusion model that jointly denoises the atomic coordinates, atom types, and lattice of a crystal, and is trained on a large database of stable inorganic compounds. Conditioning is achieved by fine-tuning a property-embedding adapter so that classifier-free guidance biases sampling toward the target values, with the guidance strength w trading off diversity against target hit rate. This is the mechanism we exploit to embed both the catalytic and *economic* objectives directly into generation.

2.4 Machine-Learning Interatomic Potentials

Evaluating surface free energies with DFT for thousands of candidates is infeasible. This work uses the universal machine-learning interatomic potential (MLIP) Universal Model for Atoms (UMA) [2] in its `fairchem` implementation. An MLIP takes an atomic configuration (atomic numbers $\{Z_i\}$ and coordinates $\{\mathbf{r}_i\}$) as input and learns a function that approximates the total potential energy $E(\{Z_i, \mathbf{r}_i\})$ of the system and the force $\mathbf{F}_i = -\partial E / \partial \mathbf{r}_i$ acting on each atom, by regressing against large-scale (structure, energy, force) data computed with DFT.

Operating principle. UMA is an *equivariant graph neural network*. Atoms are placed as graph nodes and neighbors within a cutoff radius as edges; through several layers of *message passing*, the local chemical environment of each atom (neighboring elements, distances, and angles) is encoded into a vector representation, which is then mapped to a per-atom energy contribution and summed to obtain the total energy. The physical symmetries under rotation, translation, and permutation are built into the architecture (the energy is invariant, the forces are equivariant), giving physically consistent predictions. Because the forces are obtained as the analytic gradient of the energy, an MLIP is not merely an energy predictor but provides a complete potential-energy surface that can drive *local structural relaxation* (BFGS/LBFGS, etc.)—which is central to the search for surface active sites (Section 3.3).

Transferability and role. The catalytic behavior of UMA originates from training on the vast set of adsorbate-slab calculations in the Open Catalyst (OC20, etc.) dataset [3], so that it *transfers* even to compositions and surfaces absent from training. As a result, it provides adsorption energies close to DFT at a cost 10^3 – 10^6 times lower, serving as an intermediate-accuracy step between the cheap compositional proxy and full DFT. However, for chemistries far from the training distribution (e.g., rare-earth carbide surfaces) or at weakly binding sites, its reliability may decline, so the final verdict is validated with DFT (Chapter 4).

2.5 Density-Functional Theory and Pseudopotentials

The final validation is performed with Kohn–Sham DFT using the PBE exchange-correlation functional [6] as implemented in the plane-wave code QUANTUM ESPRESSO [4], with pseudopotentials taken from the Standard Solid-State Pseudopotential (SSSP) library [5]. Where needed, the configurational disorder of the high-entropy sublattice is represented by special quasirandom structures (SQS) [15].

Chapter 3

Computational Methods

The discovery pipeline is a five-stage funnel of increasing accuracy and cost (Figure 4.1a). Low-accuracy, cheap stages filter candidates in bulk, so that accurate but expensive DFT is applied to only a few candidates. The overall flow, the tools used at each stage, and the internal procedure of the surface active-site search within Stage C are depicted in Figure 3.1. Each stage is described below.

3.1 Stage A — Data Selection, Fine-Tuning, and Conditional Generation

Stage A fine-tunes the pretrained MATTERGEN into the carbide domain to build the conditional generative model CARBIDEMATTERGEN, and then guides it toward the target properties to sample candidates. The flow is *data selection* $\rightarrow$ *labeling* $\rightarrow$ *adapter fine-tuning* $\rightarrow$ *conditional generation*.

Training-data selection. The pretrained MATTERGEN learns the distribution of stable inorganic crystals in the Alexandria–Materials Project combined dataset `alex_mp_20` [1]. From this source we select only *pure transition-metal carbides*: compositions that contain carbon (C) and at least one transition metal but *no competing anions* such as oxygen or nitrogen, thereby excluding oxycarbides and carbonitrides. This ensures that the model learns *carbide* chemistry rather than mixed-anion chemistry, ultimately yielding **5,275 training + 577 validation** structures.

Conditioning properties and labels. For each selected structure, nine conditioning properties are computed and assigned as labels (Table 3.1). For stability (`energy_above_hull`) and metallicity (`dft_band_gap`, driven to zero) we use the MATTERGEN source labels, while the remaining activity, selectivity, high-entropy, and economy properties are computed in Campaign A from literature-based *compositional proxies* (Campaign B replaces these with UMA-computed *surface* labels, Section 4.11). In particular, economy is embedded directly into the generation target by driving `expensive_metal_fraction` (the fraction of platinum-group metals Ru/Rh/Pd/Os/Ir/Pt or heavy rare earths) to zero.

Adapter fine-tuning — CarbideMatterGen. We fine-tune by attaching an embedding *adapter* for the nine properties (`property_embeddings_adapt`) to the pretrained `mattergen_base` checkpoint: each property value is embedded by a two-layer MLP, summed with the noise-step encoding, and injected into every block of the denoiser; and for classifier-free guidance, each property head is masked with probability 0.1 during training. Training uses the MATTERGEN default score-matching loss (atomic positions, lattice, and atom types) on a single

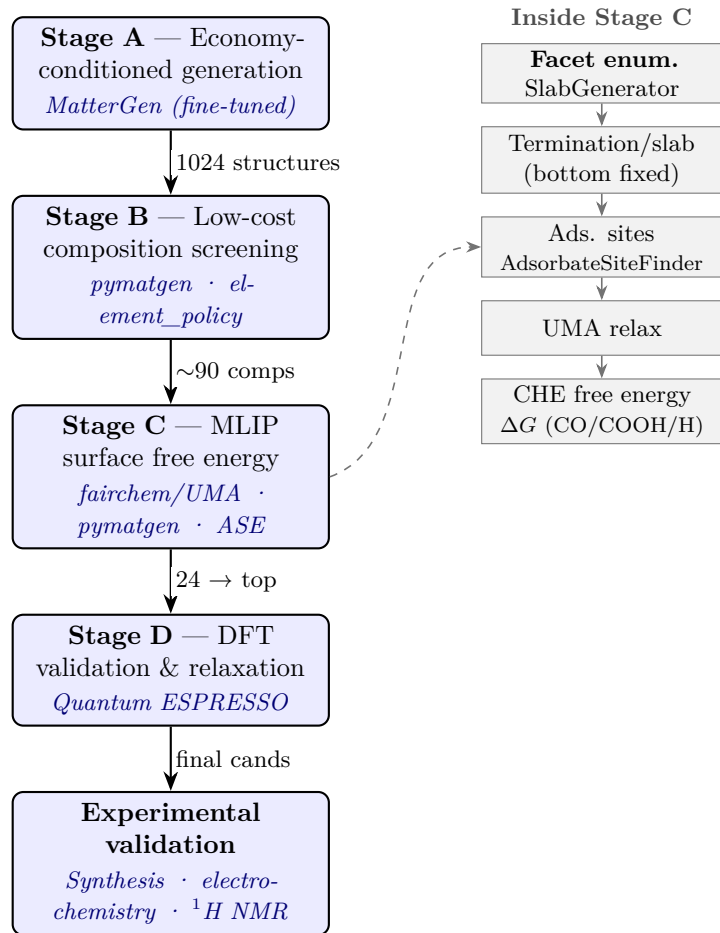


Figure 3.1: The discovery pipeline and the software used at each stage. The left-hand vertical flow is a five-stage funnel of increasing accuracy and cost (with candidate counts on each arrow). The right-hand column is the internal procedure of the surface active-site search in Stage C—low-index crystal facets are enumerated (SlabGenerator), slabs are built from each facet’s termination with the bottom fixed, and then $\ast\text{CO}/\ast\text{COOH}/\ast\text{H}$ are placed on the symmetry-distinct adsorption sites (AdsorbateSiteFinder) and relaxed with UMA to obtain the CHE free energies $\Delta G_{\text{CO}\ast}, \Delta G_{\text{COOH}\ast}, \Delta G_{\text{H}\ast}$. The two campaigns (A: activity-first baseline; B: four axioms + CO/H dual target) drive *this same* pipeline with different conditioning and constraints.

GPU with a learning rate of 5×10^{-5} , batch size 16, and up to 600 epochs, and we confirmed the absence of overfitting, as the validation loss closely tracks the training loss (Figure 4.2). We call this fine-tuned model **CarbideMatterGen**.

Conditional generation. From CARBIDEMATTERGEN, we sample a total of **1,024** candidates under the target conditioning vector at two guidance strengths ($w = 1.0$, $w = 2.5$), 512 each (per weight: 4 chunks $\times$ batch 8 $\times$ size 16). A low w emphasizes diversity, while a high w emphasizes conditioning fidelity (Section 4.2). Stages B–D then filter this candidate pool.

3.2 Stage B — Low-Cost Compositional Screening

All generated structures are re-evaluated with compositional proxies and filtered to retain those that are (i) genuine transition-metal carbides, (ii) within the CO-release region, (iii) high-entropy ($n_{\text{metal}} \geq 3$, mixing entropy $\geq 1.0 k_{\text{B}}$), and (iv) economically clean (zero fraction of expensive metals). This narrowed the candidate pool to 104 (90 unique compositions).

Table 3.1: The nine conditioning properties of CARBIDEMATTERGEN (Campaign A). “Drive” denotes the target value at generation.

Category	Property	Role/target
Stability	<code>energy_above_hull</code>	Synthesizability (small)
Metallicity	<code>dft_band_gap</code>	Conductivity ($\rightarrow 0$)
Activity	<code>best_site_co_affinity</code>	Optimal-site CO affinity ($\Delta G_{\text{CO}^*} \approx 0$)
Activity	<code>frac_co_selective_sites</code>	Fraction of CO-selective sites (large)
Activity	<code>mean_cooH_affinity</code>	CO ₂ activation (*COOH)
Selectivity	<code>co2rr_selectivity</code>	CO-vs-H selectivity
High-entropy	<code>metal_mixing_entropy</code>	Mixing entropy (large)
Packing	<code>metal_packing_density</code>	Metal-packing proxy
Economy	<code>expensive_metal_fraction</code>	Platinum-group/heavy-rare-earth fraction ($\rightarrow 0$)

3.3 Stage C — MLIP Surface Free-Energy Screening

This stage uses the MLIP to search, for each candidate composition, “which atomic site on which crystal facet is active for CO₂RR.” Because a real catalyst particle exposes multiple crystal facets and a high-entropy surface has a variety of local adsorption environments, we do not assume a particular facet or site but rather *systematically enumerate* them and compute the adsorption free energy of each. The procedure comprises the following five steps (Figure 3.1).

1. **Bulk relaxation.** The bulk cell of each composition is relaxed with UMA in both cell and coordinates simultaneously (FrechetCellFilter), to obtain a fair reference structure before slab cleaving.
2. **Facet enumeration and slab construction.** With pymatgen’s SlabGenerator, we generate the symmetrically distinct low-index facets up to maximum Miller index 1, $\{(100), (110), (111), \dots\}$. Because each facet can have different surface *terminations* (at which atomic layer it is cut), we enumerate all of them, insert a sufficient vacuum layer, then *fix the atoms of the lower half of the slab* (mimicking the bulk) and relax the upper half with UMA.
3. **Adsorption-site enumeration.** With pymatgen’s AdsorbateSiteFinder, we automatically detect the symmetrically distinct adsorption sites (ontop/bridge/hollow) on each facet, and place the three adsorbates *CO, *COOH, and *H at each site in turn to create adsorbate slabs (adslabs).
4. **MLIP relaxation.** All adslabs are locally relaxed with the UMA potential [2] until the forces fall below the threshold (Section 2.4). Because force-based relaxation is possible, the adsorbates relax from their initial placement to their actual binding geometry.
5. **Free energies and aggregation.** The adsorption free energy of each site is computed with the computational hydrogen electrode (CHE, Chapter 2):

$$\begin{aligned}
 \Delta G_{\text{CO}^*} &= E_{*\text{CO}} - E_* - E_{\text{CO(g)}} + 0.10 \text{ eV}, \\
 \Delta G_{\text{COOH}^*} &= E_{*\text{COOH}} - E_* - E_{\text{CO}_2(\text{g})} - \frac{1}{2}E_{\text{H}_2(\text{g})} + 0.41 \text{ eV}, \\
 \Delta G_{\text{H}^*} &= E_{*\text{H}} - E_* - \frac{1}{2}E_{\text{H}_2(\text{g})} + 0.24 \text{ eV},
 \end{aligned} \tag{3.1}$$

where E_* is the clean slab, $E_{(\text{g})}$ is the gas-phase molecular energy, and the constant terms are the zero-point-vibrational and entropic corrections. The clean slab and the gas-phase references are computed consistently as *total energies of the same potential* so that the slab contribution cancels in the difference. Unphysical or unconverged sites ($|\Delta G| > 10 \text{ eV}$) are excluded.

The descriptors of each candidate are aggregated across *all computed facets and sites*. Because activity can be dominated by a few optimal sites (the sites whose ΔG_{CO^*} is closest to zero), candidates are ranked by the proximity of the optimal site to the CO optimum and by the density of CO-selective sites. In the dual target of Campaign B (Section 4.11), we judge *facet by*

Table 3.2: Software and its role at each pipeline stage.

Stage	Tool	Role
Stage A	MATTERGEN (fine-tuned)	Property-conditioned crystal-structure generation
Stage B	<code>pymatgen</code> · <code>element_policy</code>	Compositional screening / element whitelist
Stage C	<code>fairchem</code> /UMA · <code>pymatgen</code> · ASE	Facet/site enumeration, MLIP relaxation, CHE ΔG
Stage D	QUANTUM ESPRESSO · ASE	DFT relaxation/SCF/DOS validation
Stability	<code>pymatgen</code> PourbaixDiagram	Bulk/surface aqueous stability

facet whether CO evolution and H repulsion hold on the *same facet*; here ΔG_{H^*} is defined as the minimum over physisorption sites, excluding dissociative and subsurface sites ($|\Delta G_{H^*}| > 1.5 \text{ eV}$) (because the global minimum is misleadingly set by a single dissociative site).

3.4 Stage D — DFT Validation and Electronic Structure

The top six candidates were validated with QUANTUM ESPRESSO [4]: a variable-cell relaxation, followed by a self-consistent-field (SCF) calculation on the relaxed structure, and then a density-of-states (DOS) calculation. We used plane-wave cutoffs of 60 Ry for the wavefunction and 480 Ry for the charge density, Marzari–Vanderbilt cold smearing (0.02 Ry), and a Monkhorst–Pack k -grid of density $\sim 30 \text{ \AA}/a$. For C, Y, Zr, and W we used SSSP pseudopotentials [5], and for the rare earths (La, Ce, Pr) norm-conserving pseudopotentials with the $4f$ shell frozen into the core (trivalent), which is physically appropriate for the localized $4f$ states of these metallic carbides. The element-projected DOS was obtained by projecting the converged Kohn–Sham states onto atomic orbitals (with no additional SCF), and summing per element to quantify the extent to which each sublattice contributes to the states at the Fermi level.

3.5 Stage A' — Solid-Solution Supercells

To represent explicit chemical disorder, we constructed special quasirandom structures [15] of the high-entropy shortlist on a rocksalt (MC) or hexagonal (M_2C) framework by Monte Carlo matching of the pair-correlation functions of the metal sublattice. These supercells are the substrate for future studies of explicit disordered surfaces.

3.6 Software

The pipeline combines open-source tools at each stage (Table 3.2). Generation is handled by the fine-tuned diffusion model MATTERGEN [1], MLIP surface screening by the UMA implementation in `fairchem` [2], while slab and surface enumeration, adsorption-site detection, structure processing, and bulk and surface Pourbaix analysis are handled by `pymatgen` [16], and atomic-structure I/O and the relaxation drivers by the Atomic Simulation Environment (ASE) [17]. The final DFT validation is performed with GPU-accelerated QUANTUM ESPRESSO [4]. The element-palette whitelist (`element_policy`), the aggregation of compositional/surface descriptors, and the dual-target judgment were implemented as scripts accompanying this work.

Chapter 4

Computational Results

An outstanding CO₂RR carbide electrocatalyst must *simultaneously* satisfy four design axioms: (A1) **activity** — activate CO₂ and release CO (the *gateway intermediate* of all more deeply reduced products) near thermoneutrality ($\Delta G_{\text{CO}^*} \approx 0$); (A2) **selectivity** — suppress the competing hydrogen evolution reaction (HER), i.e. the surface repels H ($\Delta G_{\text{H}^*} > 0$) so that the active site is not poisoned by *H; (A3) **manufacturability** — survival of the leaching step of the metallothermic-reduction-acid-leaching synthesis route; and (A4) **economy** — free of platinum-group and precious metals. This chapter drives a single generation pipeline with *two distinct constraint sets*, using a controlled comparison to elucidate how these axioms interact at the generation stage.

Campaign A (activity-first baseline; Sections 4.1–4.9) conditions only on activity (A1), stability, and economy (A4), imposing neither selectivity (A2) nor manufacturability (A3). This campaign both demonstrates that the pipeline works and reveals *where an unconstrained activity-first search converges* — the result is light-rare-earth-rich high-entropy carbides. The *manufacturability gate (Section 4.10)* shows that this candidate pool fails at axiom A3: rare earths dissolve in the indispensable acid-leaching step. *Campaign B (fully constrained dual objective; Section 4.11)* imposes all four axioms — in particular, it makes selectivity (A2) explicit as a CO/H *dual active-site* criterion and restricts the search space to acid-stable transition metals — and derives three Ti-based transition-metal carbides. The only difference between the two campaigns is the *constraint set*, and that contrast is the central argument of this work.

4.1 Campaign A — Generation and the Discovery Funnel

Fine-tuning converged smoothly (Figure 4.2), and conditional generation produced 1024 structures, of which 1004 are transition-metal carbides. Their composition–descriptor distribution (Figure 4.3) shows that the economy conditioning was effective: the fraction of expensive metals is essentially zero across the candidate pool, and the metal palette is dominated by inexpensive early transition metals and light rare earths (W, Y, Zr, Ce, La, Pr). Low-cost screening preserved 104 economical CO-releasing high-entropy carbides. The full funnel (1024 → 1004 → 90 → 24 → 6) is summarized in Figure 4.1a.

4.2 MLIP Surface Screening

UMA surface screening evaluated all low-index facets of the 24 candidates (Figure 4.4). The best-site ΔG_{CO^*} of the leading candidates lies within tens of meV of the CO optimum, and the best site is found not on a single preferred facet but across a variety of facets. This is consistent with the high-entropy design intent of dispersing favorable sites. The top six candidates, all CO-over-H selective at their best site, are collected in Table 4.1.

Table 4.1: Master summary of the six final economical high-entropy carbide candidates: the surface CO-release descriptor (best-site ΔG_{CO^*} from UMA screening), CO-over-H selectivity, and the DFT-validated Fermi energy, density of states at E_F , dominant conduction element, and residual pressure. All six are metallic.

Candidate	ΔG_{CO^*} (eV)	CO>H	E_F (eV)	DOS(E_F)	Dominant element	Metallic	P (kbar)
$\text{Pr}_3\text{Y}_2\text{WC}_7$	−0.001	Yes	12.00	7.4	C	Yes	+0.15
$\text{La}_5\text{Y}_4\text{W}_2\text{C}_9$	−0.001	Yes	11.52	9.8	C	Yes	−0.10
$\text{Ce}_2\text{Y}_2\text{ZrWC}_6$	+0.007	Yes	12.72	6.7	C	Yes	−0.29
$\text{La}_3\text{Y}_5(\text{WC}_3)_3$	−0.008	Yes	12.24	12.4	Y	Yes	−0.32
Y_2ZrWC_8	−0.010	Yes	10.52	1.7	W	Yes	+0.41
$\text{CeY}_2\text{ZrWC}_7$	−0.011	Yes	12.85	3.3	C	Yes	−0.13

4.3 DFT Validation: Stability and Metallicity

All six candidates relaxed with low residual pressure ($|P| < 0.5 \text{ kbar}$), and they are *metals* with a finite density of states at the Fermi level ranging over 1.7–12.4 states/eV per cell (Figure 4.6, Table 4.1). Metallic conductivity, a prerequisite for an electrocatalyst, is confirmed from first principles.

4.4 Conduction Character from Projected DOS

The element-projected DOS (Figure 4.5) identifies *which* sublattice carries the states at the Fermi level. Two regimes appear: the carbon-rich candidates ($\text{Pr}_3\text{Y}_2\text{WC}_7$, $\text{Ce}_2\text{Y}_2\text{ZrWC}_6$) are dominated at E_F by carbon $2p$ character ($\sim 60\%$), indicating a conductive carbide network, whereas the tungsten-rich $\text{La}_3\text{Y}_5(\text{WC}_3)_3$ and the rare-earth-free Y_2ZrWC_8 are dominated by metal d (Y/La/W) states. The element-wise weights are plotted in Figure 4.1d. This C- p /metal- d hybridized conduction matches exactly the electronic signature expected of “platinum-like” carbide catalysts [12].

4.5 Master Table

Table 4.1 synthesizes the surface activity descriptors and DFT validation of the six final candidates; the corresponding synthesis figure is Figure 4.1.

4.6 Surface Free Energies and CO_2RR Activity (DFT)

For the SQS supercells of the six candidates, we enumerated the symmetrically distinct low-index facets of each crystal structure, selected the most stable termination for each facet, and then placed $\ast\text{CO}/\ast\text{COOH}/\ast\text{H}$ on every site of the three most active facets and relaxed them; the resulting geometries (Stage B, UMA screen) were recomputed as DFT (QE) single-points. In all, the DFT surface free energies ΔG_{CO^*} , ΔG_{COOH^*} , ΔG_{H^*} of **108 slabs** (6 candidates $\times$ 3 facets $\times$ [clean + 3 adsorbates], with gas references $\text{CO}/\text{CO}_2/\text{H}_2$) were completed (for the computational protocol see Chapter 3: DFT//UMA single-points, Γ point, $\text{ecut } 60/480 \text{ Ry}$, MV smearing).

DFT validation of the MLIP screen. The DFT and UMA (MLIP) free energies correlate strongly across all 54 (facet, adsorbate) pairs with Pearson $r = 0.973$ (MAE 0.342 eV, RMSE 0.457 eV), with adsorbate-wise MAEs of CO 0.24 / H 0.30 / COOH 0.49 eV. That is, the *MLIP-level activity ranking is reproduced at the DFT level*. The error is largest for COOH because of

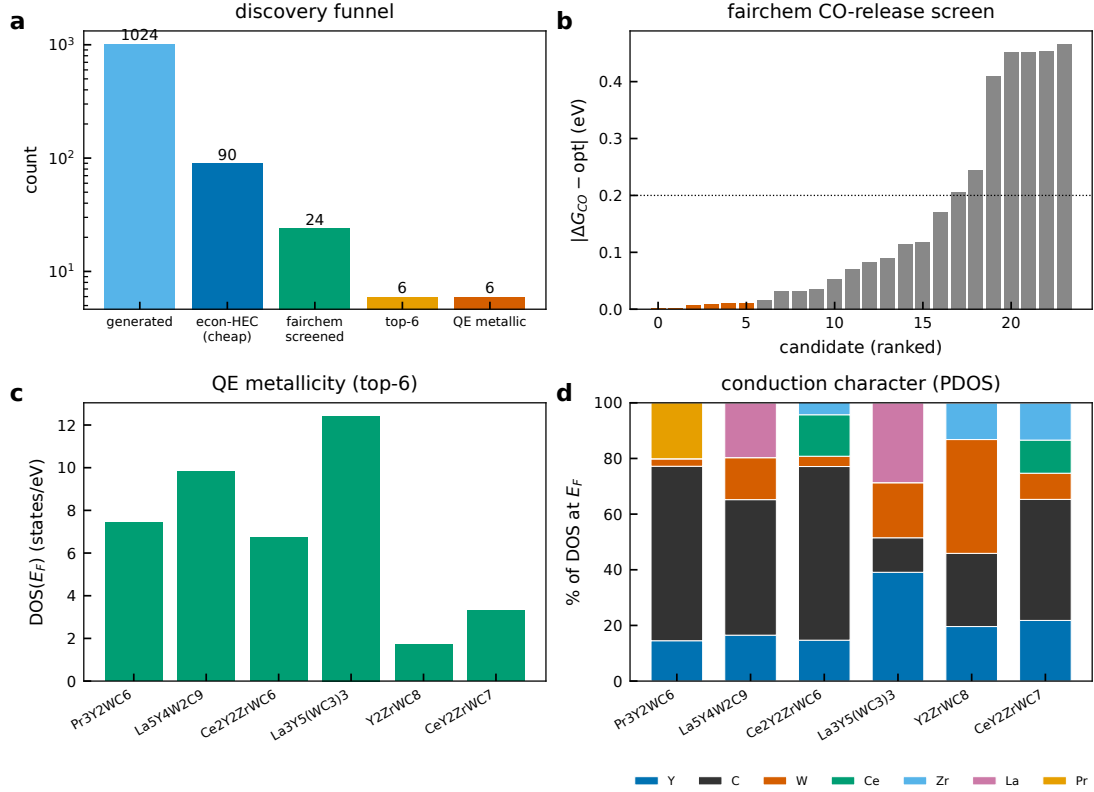


Figure 4.1: End-to-end discovery synthesis. (a) Discovery funnel from 1024 generated structures to 6 DFT-validated metallic candidates. (b) UMA surface CO-release descriptor of the screened candidates (top six highlighted). (c) Fermi-level DFT density of states of the top six. (d) Element-projected conduction character (element-wise DOS weight at E_F).

sites at certain low-symmetry facets where $\ast\text{COOH}$ overbinds dissociatively ($\Delta G_{\text{COOH}\ast} \lesssim -2$ eV, Table 4.2), and DFT partially corrects this unphysical deep well.

Computational hydrogen electrode (CHE) free-energy diagram and limiting potential. For the two-electron $\text{CO}_2 \rightarrow \text{CO}$ path $\text{CO}_2 + \ast \rightarrow \ast\text{COOH} \rightarrow \ast\text{CO} \rightarrow \text{CO}(\text{g}) + \ast$, from the two proton-electron coupled transfer steps $\Delta G_1 = \Delta G_{\text{COOH}\ast}$ and $\Delta G_2 = \Delta G_{\text{CO}\ast} - \Delta G_{\text{COOH}\ast}$ we define the limiting potential $U_L = -\max(\Delta G_1, \Delta G_2)/e$ (Figure 4.7). **All six candidates have $\Delta G_{\text{COOH}\ast} < \Delta G_{\text{H}\ast}$ on their best facet**, i.e. CO_2 activation is favored over the competing hydrogen evolution reaction (HER), confirming CO selectivity at the DFT level (Table 4.2). The limiting potentials, however, split clearly into two groups: $\text{CeY}_2\text{ZrWC}_7$ ($U_L = -0.10$ V) and $\text{Pr}_3\text{Y}_2\text{WC}_6$ ($U_L = -0.21$ V) have a well-defined molecular intermediate that binds $\ast\text{COOH}$ moderately and thus a low overpotential, whereas the remaining four overbind $\ast\text{COOH}$ (down to -7.8 eV for $\text{La}_5\text{Y}_4\text{W}_2\text{C}_9$), making the $\ast\text{COOH} \rightarrow \ast\text{CO}$ step rate-limiting and giving a large apparent overpotential. For all candidates the potential-determining step (PDS) was the second reduction step ($\ast\text{COOH} \rightarrow \ast\text{CO}$). DFT therefore identifies $\text{CeY}_2\text{ZrWC}_7$ and $\text{Pr}_3\text{Y}_2\text{WC}_6$ as *low-overpotential CO_2RR candidates*.

d -band center and adsorption descriptor. By projecting each candidate’s stored Kohn-Sham wavefunctions onto atomic orbitals (projwfc), we obtained the d -band center ε_d (relative to E_F) from the transition-metal d -projected density of states (Figure 4.8). The metal-averaged ε_d spans from $\text{Pr}_3\text{Y}_2\text{WC}_6$ (-0.48 eV) to Y_2ZrWC_8 (-1.46 eV), and element-wise, W has the deepest center (-1 to -2.3 eV) while the rare-earth $5d$ and Zr are closer to E_F . Consistent with

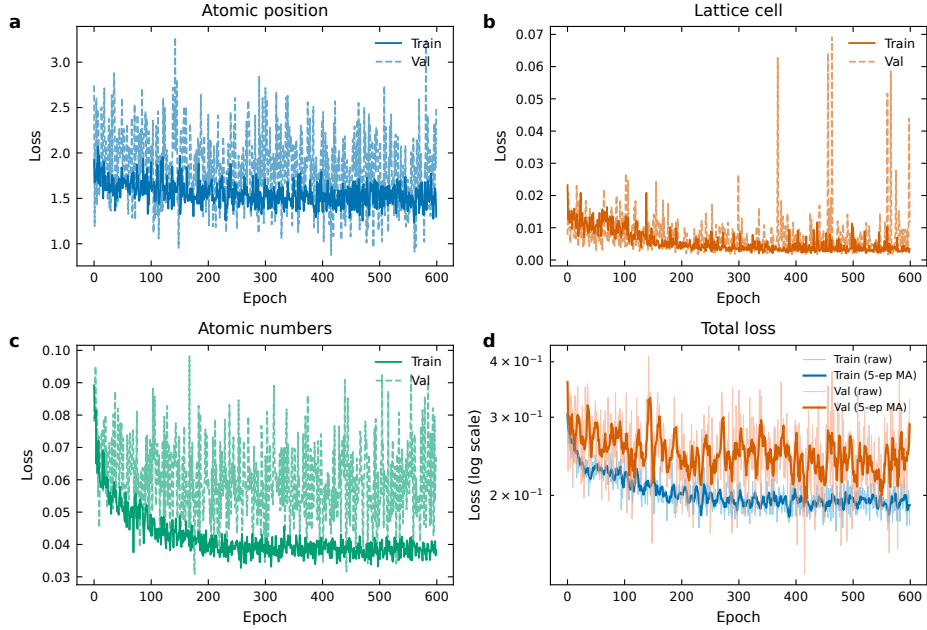


Figure 4.2: Fine-tuning of the economy-conditioned generation model: training/validation loss over 600 epochs and the component-wise (position, lattice, atom type) losses.

Table 4.2: DFT surface descriptors and CHE limiting potential of each candidate’s best facet. ΔG in eV, U_L in V vs RHE. All candidates have $\Delta G_{\text{COOH}^*} < \Delta G_{\text{H}^*}$ (CO-selective). PDS is the potential-determining step.

Candidate	Best facet	ΔG_{COOH^*}	ΔG_{CO^*}	ΔG_{H^*}	$U_L(\text{DFT})$	$U_L(\text{UMA})$
CeY ₂ ZrWC ₇	(011)	−0.47	−0.37	−0.42	−0.10	−1.80
Pr ₃ Y ₂ WC ₆	(010)	−0.63	−0.42	−0.23	−0.21	−0.66
La ₃ Y ₅ (WC ₃) ₃	(101)	−1.28	+0.13	−0.34	−1.40	−2.19
Ce ₂ Y ₂ ZrWC ₆	(101)	−3.12	−0.81	−1.38	−2.31	−2.99
La ₅ Y ₄ W ₂ C ₉	(111)	−2.40	−0.03	−0.04	−2.37	−2.95
Y ₂ ZrWC ₈	(100)	−2.19	+0.22	−0.30	−2.41	−2.50

the d -band model, candidates whose center is closer to E_F (higher ε_d) tend to bind CO more strongly, linking the conduction character to the adsorption descriptor.

4.7 Formation Energies and Convex-Hull Stability (DFT + Materials Project)

The bulk formation energies of the six candidates were computed from elemental references at the same QE-PBE level (W bcc, Zr/Y hcp, La/Ce/Pr fcc, C diamond; vc-relax), and the convex hull was constructed from competing phases in the same chemical space (Materials Project GGA entries, including all subsystems) to evaluate the above-hull energy E_{hull} (Table 4.3). Owing to elemental-reference cancellation, the difference between the QE and MP (VASP) formation energies is largely removed, and the residual is estimated at the ~ 0.1 eV/atom level. The E_{hull} of the six candidates is +0.15 to +0.36 eV/atom, which is *metastable* on a 0 K enthalpy basis — typical of high-entropy carbides (entropy-stabilized materials, not enthalpic ground states). The stabilization from the ideal configurational entropy, $-T\Delta S_{\text{mix}}$, is −0.05 to −0.10 eV/atom at 1800 K, which reduces the hull distance but by itself does not fully cancel it (crossover temperature $T^* \approx 3000$ to 12000 K); the actual synthesizability therefore depends jointly on vibrational entropy, non-ideal mixing enthalpy, *kinetic trapping* by high-temperature quenching, and the

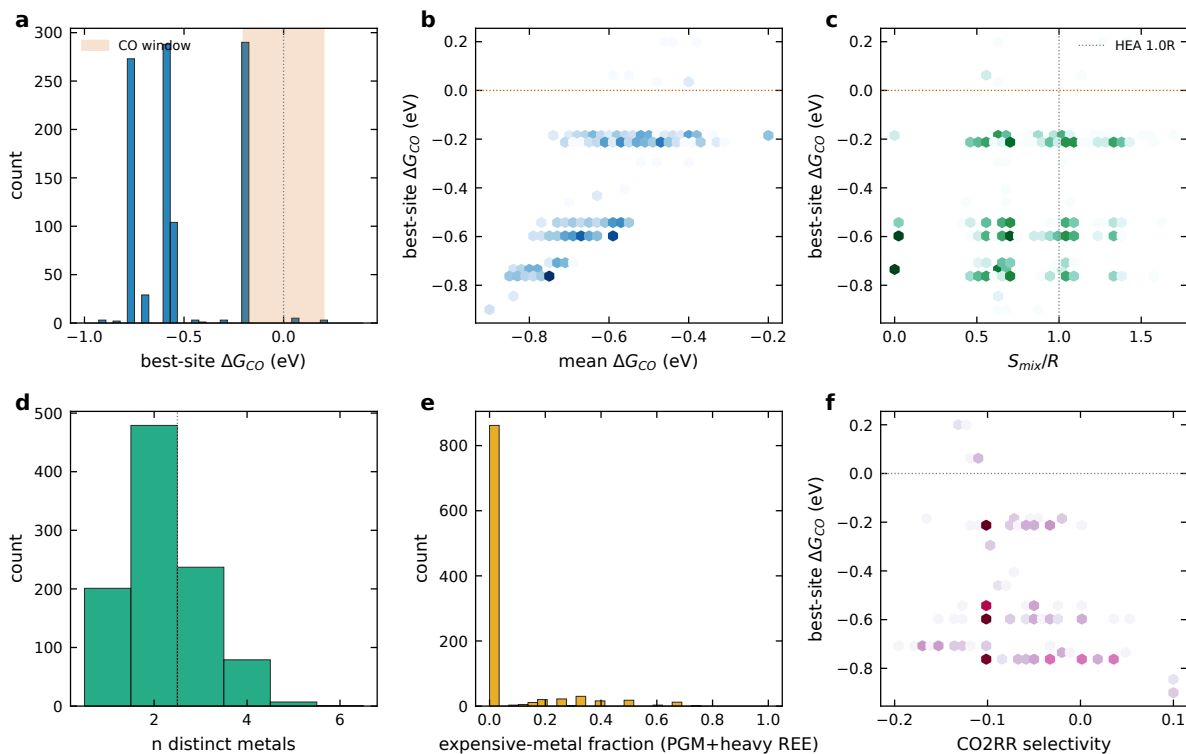


Figure 4.3: Composition statistics of the generated carbide candidate pool: (a) best-site CO affinity shown together with the CO-release window; (b,c) descriptor correlations; (d) distribution of metal count; (e) fraction of expensive metals (economy constraint); (f) CO₂RR selectivity.

methodological uncertainties above. In short, these ML-generated candidates are metastable phases whose synthesizability is not self-evident, which directly motivates the synthesis and phase-purity tests (XRD/XPS) of Chapter 5.

4.8 Refined Validation (Stage C): DFT Relaxation and k -Convergence of Top Candidates

The surface free energies presented so far are DFT single-points on UMA-relaxed geometries (DFT//UMA, Γ). For the two most active candidates (CeY₂ZrWC₇, Pr₃Y₂WC₆), we refined the headline results by *full ionic relaxation* (GPU QE, bottom half fixed, Γ) of the clean slab, *CO, *COOH, and *H on their active facet.

k -point convergence. The symmetry-free ~ 50 -atom slabs cannot run with $k > \Gamma$ on 16 GB of VRAM, but ΔG is a difference at the same cell and same k , so finite- k errors cancel. On a 36-atom slab, the Γ vs $2 \times 1 \times 1$ difference in adsorption free energy is ΔG_{CO} 0.01 eV and ΔG_{COOH} 0.11 eV; even though the absolute slab energy shifts by ~ 0.7 eV, the difference is stable, justifying the Γ -level ΔG .

Relaxation refines the top ranking. Full relaxation confirmed that **the low overpotential of CeY₂ZrWC₇ is robust**: $\Delta G_{\text{CO}} = -0.09$ eV (nearly perfect thermoneutrality; improved by relaxation from the single-point -0.37), $U_L = -0.15$ V (a 0.05 V shift from the single-point -0.10). By contrast, **the single-point activity of Pr₃Y₂WC₆ was not robust**: upon relaxation the *COOH on the active facet *dissociates* (in the relaxed geometry one C–O bond breaks to 3.3 Å, decomposing into *CO+*O), falling into a deep well of $\Delta G_{\text{COOH}} = -3.66$ eV and

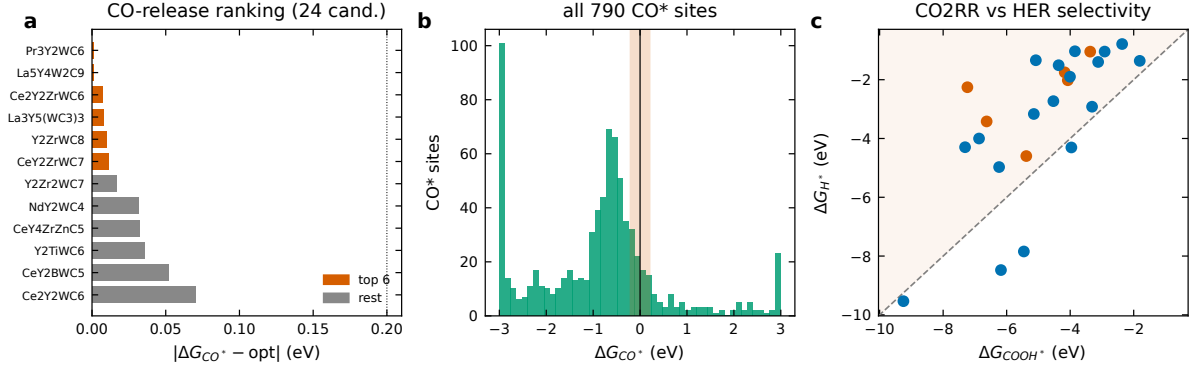


Figure 4.4: UMA surface screening: (a) candidates ranked by the proximity of their best site to the CO optimum; (b) aggregated ΔG_{CO^*} distribution shown together with the CO-release window; (c) CO-over-H₂ selectivity (ΔG_{COH^*} vs ΔG_{H^*}).

Table 4.3: DFT formation energies (QE elemental references), above-hull energy relative to the MP convex hull, and ideal configurational-entropy stabilization. E_f , E_{hull} in eV/atom. T^* is the temperature at which the ideal entropy cancels the hull distance.

Candidate	E_f (QE)	E_{hull}	$-T\Delta S_{\text{mix}}(1800 \text{ K})$	$T^*(\text{K})$
La ₃ Y ₅ (WC ₃) ₃	-0.087	+0.154	-0.091	3050
La ₅ Y ₄ W ₂ C ₉	-0.024	+0.191	-0.088	3890
CeY ₂ ZrWC ₇	-0.118	+0.223	-0.086	4660
Pr ₃ Y ₂ WC ₆	+0.046	+0.265	-0.078	6080
Ce ₂ Y ₂ ZrWC ₆	-0.067	+0.309	-0.103	5390
Y ₂ ZrWC ₈	+0.072	+0.360	-0.054	12040

worsening U_L to -3.1 V. That is, the low single-point U_L of Pr₃Y₂WC₆ was an artifact of the UMA geometry. Both candidates show a tendency for *COOH dissociation, but in CeY₂ZrWC₇ the decomposition products bind moderately, so activity is preserved. Therefore *a ΔG_{CO} that is robust to relaxation is the reliable activity descriptor*, and by this criterion **CeY₂ZrWC₇** ($\Delta G_{CO} = -0.09$ eV) **emerges as the uniquely ideal CO-evolution candidate**. The microscopic pathway and microkinetics of *COOH dissociation are subjects for future work.

4.9 Solid-Solution Supercells and Ongoing Calculations

In Stage A' (Chapter 3) the special quasirandom structures of the high-entropy shortlist were built as explicit disordered supercells on rocksalt and hexagonal frameworks (metal-sublattice pair-correlation error minimized), and they are the substrates of the surface study above (Section 4.6). Since the *final deliverables of this work are the three Ti-based rocksalt carbides that pass all four design axioms A1–A4* (hec3-MoTiZr/CrTiV/CrTiZr), we here summarize the **first-principles validation status of these three**.

Completed validation (three final carbides).

- **Dual-objective surface free energies and limiting potential:** per-facet DFT//UMA confirms simultaneous CO evolution ($|\Delta G_{CO^*}| < 0.3$) and H repulsion ($\Delta G_{H^*} > 0$) and yields U_L (Table 4.6, Figure 4.10b).
- **Full relaxation of the representative candidate:** hec3-MoTiZr (100) was fully relaxed with GPU DFT, reconfirming selectivity ($\Delta G_{H^*} = +0.57$; CO binds weakly on top Mo and is released) and correcting U_L to -0.40 V (Section 4.8, Figure 4.13).
- **Convex-hull stability:** from QE-PBE formation energies against the Materials Project

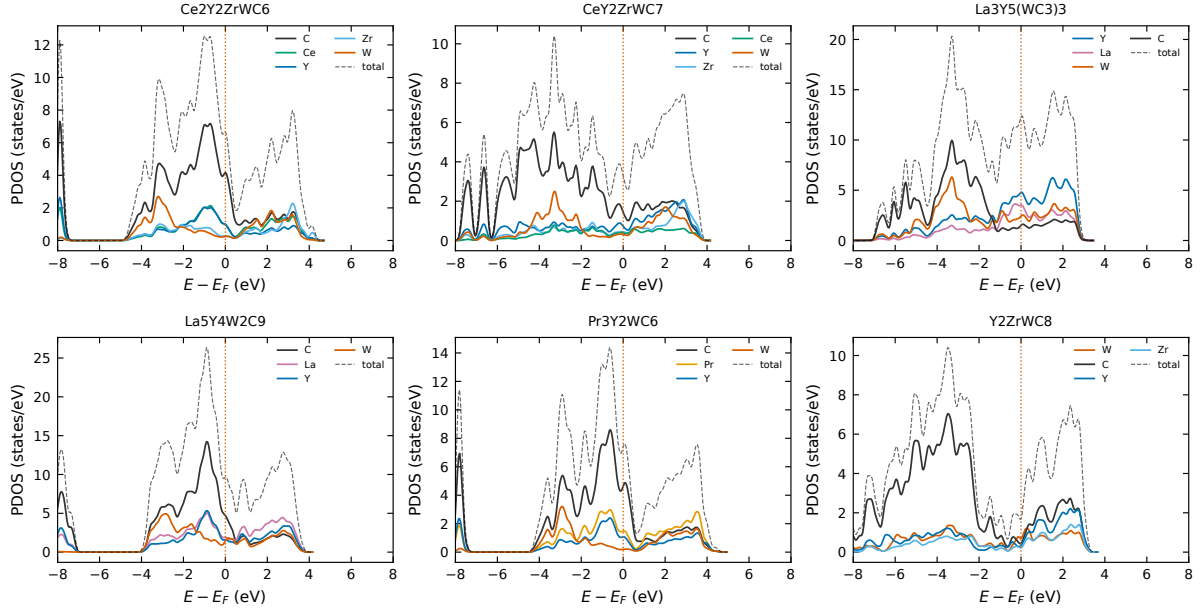


Figure 4.5: Element-projected density of states (PDOS) of the six DFT-validated candidates, referenced to $E - E_F$. The finite states at E_F (dashed line) confirm metallicity, and the element-wise decomposition identifies the conduction character.

hull, all three are more stable than Campaign A and MoTiZr sits effectively on the hull ($E_{\text{hull}} \approx -0.02$) (Table 4.5, Section 4.7).

- **Bulk metallicity and conduction character:** QE bulk DOS/PDOS show all three are metallic (DOS(E_F) 21–35 states/eV) and that the CO-binding group-6 metal (Mo/Cr) dominates the states at E_F (Figure 4.10c,d).
- **Multicarbon branching:** the three fates of *CO on the (100) surface (CO release / C1 *CHO / C2 *OCCO) were computed, and the C2 coupling is stabilized, consistent with the experimentally observed ethanol (Figure 4.12, Section 6.6).

Remaining first-principles tasks (three final carbides).

- [Full relaxation of hec3-CrTiV and CrTiZr (currently single-point) to finalize U_L – to be added when the calculation completes]
- [Coverage-dependent microkinetics with lateral interactions and a reaction-diffusion local-pH model – to be added when the calculation completes]
- [Projected crystal-orbital (COHP/COOP, LOBSTER) bond-strength analysis to quantify the group-6–CO bond – to be added when the calculation completes]
- [An operating-potential Pourbaix of the *carbide* surface (including lattice stabilization) for oxide-film / OH-poisoning margins – to be added when the calculation completes]
- [Explicit first-principles C–C coupling / ethanol pathway (transition states and barriers) – to be added when the calculation completes]

Role of the Campaign-A (rare-earth) diagnostics. Separately, for the six rare-earth candidates of Campaign A we also computed the *dissociative overbinding* of *COOH on the active facet (on $\text{Pr}_3\text{Y}_2\text{WC}_6$ one C–O bond breaks to 3.3 Å into *CO+*O, Section 4.8) and the saturated *mean-field coverage* at operating potential ($\theta_{*}\text{COOH} \approx 0.87\text{--}1.0$). These are *not* a validation of the final candidates but **counter-evidence** that the activity-first (unconstrained) search is mechanically strained as well — this candidate set is eliminated at the manufacturability

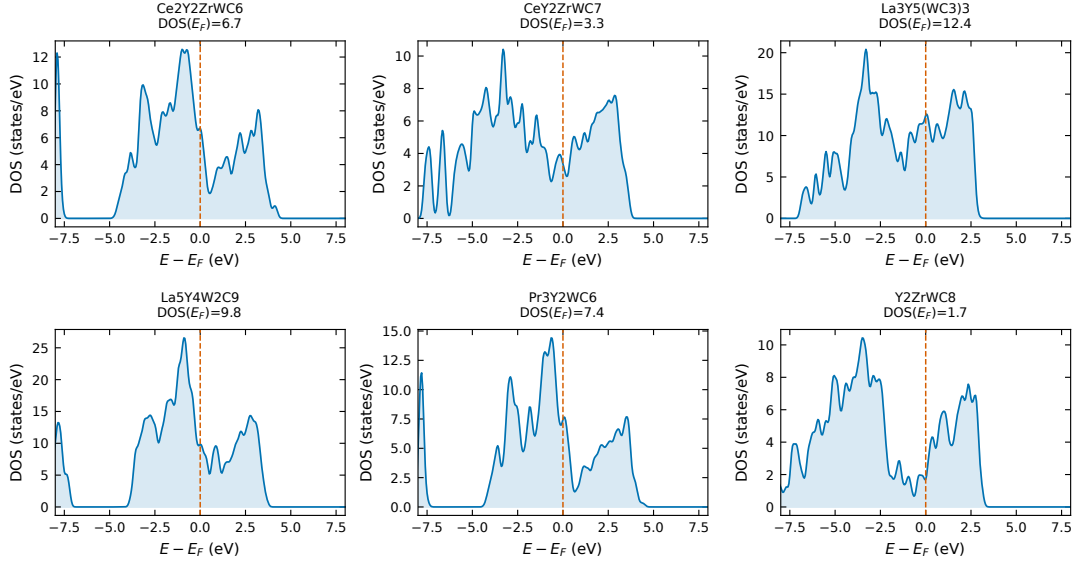


Figure 4.6: Total density of states of the six candidates near the Fermi level (DFT, PBE).

gate of the next section (Section 4.10), which justifies the manufacturability constraint (A3) of Campaign B. Further computation on the rare-earth candidates (e.g. COHP, coverage-dependent DFT) is therefore *unnecessary*; the remaining tasks focus on the three final carbides above.

4.10 Manufacturability Gate: Aqueous Oxidation/Leaching of Rare Earths and Active-Site Poisoning

The shortlist of Campaign A was selected only for activity, stability, and economy, and was not validated against design axiom **A3 (manufacturability)**. This section tests this gate falsifiably with first-principles surface thermodynamics and experimental-thermodynamics Pourbaix analysis. *To state the conclusion up front, the Campaign A candidate pool does not pass this gate* — and this fact is the basis for promoting manufacturability to a generation constraint in Campaign B. The shortlist includes light rare earths such as La, Ce, and Pr on the metal sublattice. These have very high oxygen affinity: their standard reduction potentials for $M^{3+} + 3e^- \rightarrow M$ are La -2.38 , Ce -2.34 , Pr -2.35 , Y -2.37 V (SHE), far more negative than the CO₂RR operating window (≈ -0.5 to -1.1 V vs RHE). That is, in the *metallic state* the rare earths are thermodynamically strongly inclined toward oxidation/dissolution (e.g. $M_{(aq)}^{3+}$ or $M(OH)_3/M_2O_3$). The key question is therefore “does the stabilization of the carbide lattice offset this driving force and preserve the active site,” which can be tested falsifiably at three levels of first-principles calculation.

(1) Surface Pourbaix / *O, *OH coverage (direct test of active-site poisoning). In the same slab and CHE framework as *CO/*COOH/*H, we compute the adsorption free energies of *O and *OH: $\Delta G_{*OH} = G_{*OH} - G_* - (G_{H_2O} - \frac{1}{2}G_{H_2}) + eU$, $\Delta G_{*O} = G_{*O} - G_* - (G_{H_2O} - G_{H_2}) + 2eU$. Taking, at a given (U, pH), the surface species that minimizes the free energy and constructing a *surface Pourbaix diagram* determines whether, at the operating potential, the surface is clean/*H-covered or poisoned by *OH/*O. In particular, if *OH binds more strongly than *CO/*COOH at the rare-earth site, active-site poisoning is predicted. *Mitigating factor*: CO₂RR operates at cathodic (reducing) potentials, so oxidation is thermodynamically suppressed (cathodic protection), and the poisoning risk during normal operation is low; the risk is greatest at open-circuit/startup and under local alkalization — the Pourbaix diagram

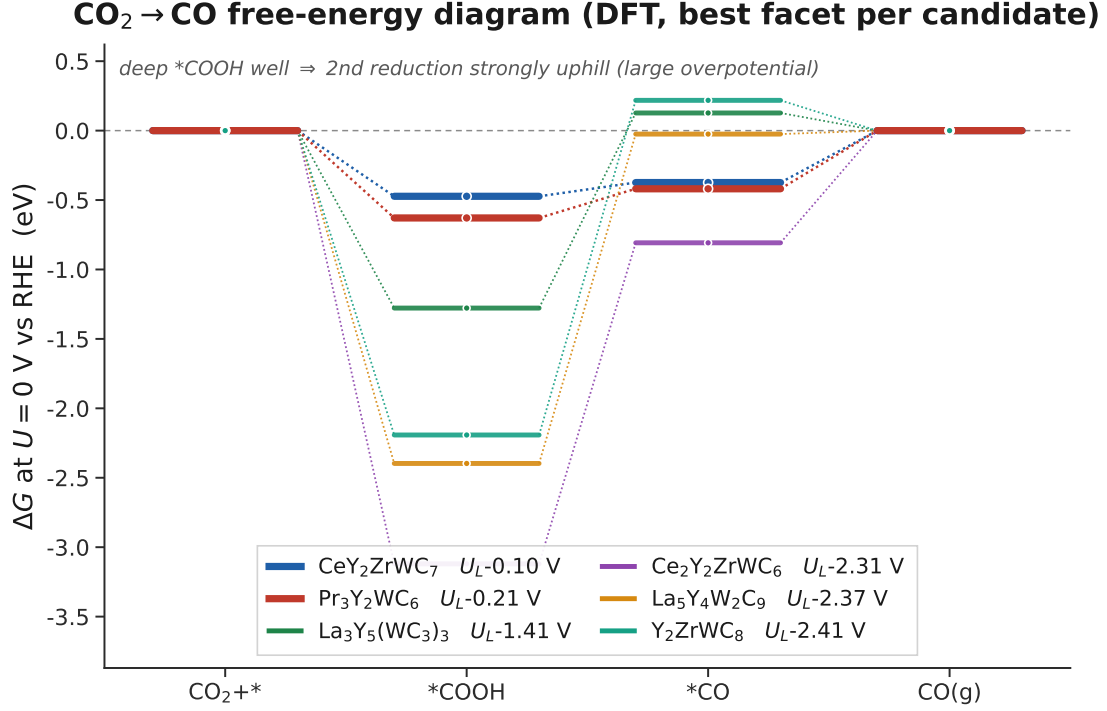


Figure 4.7: CO₂ → CO computational hydrogen electrode (CHE) free-energy diagrams of the six candidates ($U = 0$ vs RHE, best facet of each candidate). For the low-overpotential candidates (CeY₂ZrWC₇, Pr₃Y₂WC₆), *COOH is a moderate intermediate state, whereas for the others the *COOH deep well makes the second reduction strongly endothermic.

quantifies this potential/pH dependence.

**O/*OH screen (UMA).* We added *O and *OH as adsorbates to the Stage B pipeline and screened all facets and all sites of the six candidates (120–306 sites per candidate). The free energies of the strongest-binding sites are $\Delta G_{*OH}(0) = -2.4$ to -7.0 eV, $\Delta G_{*O}(0) = -1.2$ to -8.8 eV, far stronger than *CO ($\Delta G_{CO} \approx 0$), showing the large oxygen affinity of these rare-earth-carbide surfaces. This suggests a thermodynamic *tendency* toward oxidative coverage, but the steady state during actual operation must be judged by a same-facet comparison that includes the reaction intermediate *COOH (see DFT correction below), and it then turns out that *COOH prevails rather than oxidation.

Active-facet same-facet full Pourbaix (DFT, all six candidates completed). To judge the steady state rigorously, the competing adsorbates must be compared on the *same facet*, and in particular including the CO₂RR reaction intermediate *COOH ($\Delta G_{*COOH}(U) = \Delta G_{*COOH}(0) + eU$). To this end, on each candidate’s *active facet* (the facet of minimum U_L , Table 4.2) we recomputed *O/*OH with QE (24 slabs, GPU; La-rich and large slabs with a CPU fallback) and bound them on the same facet as *CO/*COOH/*H. **On the active facet the *OH/*O binding is far weaker than on the worst facet** ($\Delta G_{*OH} = -0.8$ to -1.6 eV, with some $\Delta G_{*O} \gtrsim 0$) — i.e. the actual reaction facet has low oxygen affinity. The conclusion of the six-candidate same-facet Pourbaix (Figure 4.9): **at the operating potential (-0.3 to -1.0 V vs RHE), all six candidates have the reaction intermediate *COOH as the steady state** (6/6 at or below -0.6 V, 5/6 at -0.3 V; only La₃Y₅ switches at -0.6 V), and *O/*OH oxidative coverage is confined to a *transient state near open circuit* (~ 0 V). Thus, *under cathodic operating conditions it is not oxidative poisoning but *COOH (the operating state) that dominates*, and the earlier “complete poisoning” verdict was an artifact of (i) the omission of *COOH and (ii) facet mixing. The oxidation risk is real, but its stage is *startup/open-circuit*, and the substantive axis of candidate differentiation is not oxidation but the *COOH binding strength (activity,

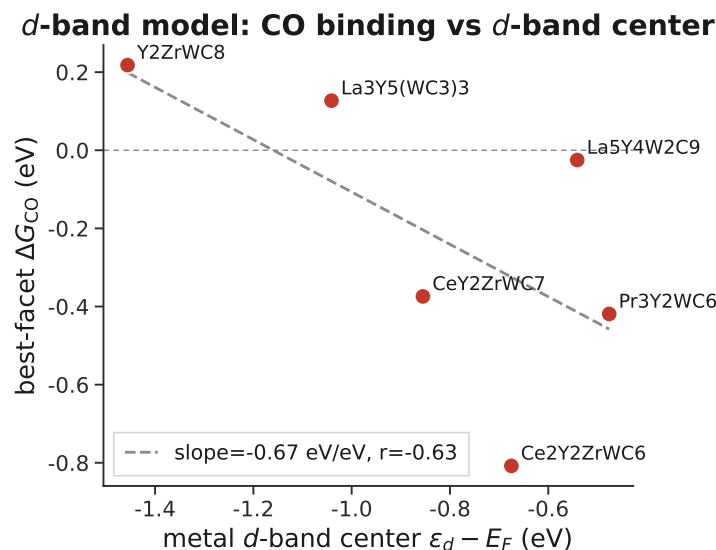


Figure 4.8: Metal d -band center $\varepsilon_d - E_F$ vs best-facet ΔG_{CO} . A higher d -band tends to give stronger CO binding (trend line), linking conduction character and adsorption.

Table 4.2). The agreement with UMA is $r = 0.74$ (MAE 0.60 eV) on the active facet (weak binding), lower than on the worst facet ($r = 0.99$), which means that MLIP reliability degrades at weakly binding sites, and DFT (the reference) actually gives *even weaker* binding (lower oxygen affinity). Cross-validation is carried out via the XPS oxidation-state and post-leaching analysis of Chapter 5 and in-situ FT-IR (*COOH/*CO bands).

This calculation reuses the Stage B/Phase A pipeline (adding *O and *OH as adsorbates; the `--adsorbates OH,0` option of the script `co_surface_screen` and `61_surface_pourbaix.py`).

(2) Bulk Pourbaix / leaching thermodynamics (completed). Whether the constituent metals remain metallic (immune) at the operating conditions, passivate as solid oxides/hydroxides, or leach as dissolved ions, was judged with *experimental-thermodynamics* Pourbaix analysis (the ΔG_f of aqueous ions and reference oxides from the Materials Project ion reference (Wagman/NBS), hydroxides from CRC/Wagman tables; Nernst potential/pH dependence computed with pymatgen PourbaixDiagram). Under CO₂RR conditions (pH 7, including local cathodic alkalization pH 10; $U = 0$ to -1.1 V vs RHE) the results are (Table 4.4): **no metal is immune (metal-stable) at the operating potential** (except W under cathodic bias). La, Ce, and Zr *passivate* as hydroxide films, Pr and Y *leach* as ions (though Y passivates at local alkaline pH), and W dissolves as tungstate (WO_4^{2-}) at open circuit but is immune under cathodic bias. That is, the bulk thermodynamics predicts *oxidation (passivation or leaching) at open-circuit/startup*, consistent with the surface result (operating potential = *COOH steady state). *Note:* this is a *pure-metal* reference and thus a *leaching upper bound* that does not include the carbide lattice stabilization (the formation energy of Section 4.7); the actual carbide is more stable. The tabulated-value uncertainty of the hydroxide ΔG_f ($\sim$ tens of kJ/mol) also has little effect on the qualitative verdict.

(3) Explicit surface oxidation / carbon-vacancy formation energies (optional). We compute, by DFT, the energy of substituting surface C with O or forming an oxygen overlayer, and the surface carbon-vacancy formation energy, to reinforce the thermodynamics of (1) with the local barrier the lattice presents against initial oxidation.

Together, these three tests elucidate, quantitatively and falsifiably, the “risk that even rare earths bound in the lattice cover the active sites by oxidation/leaching upon exposure to the

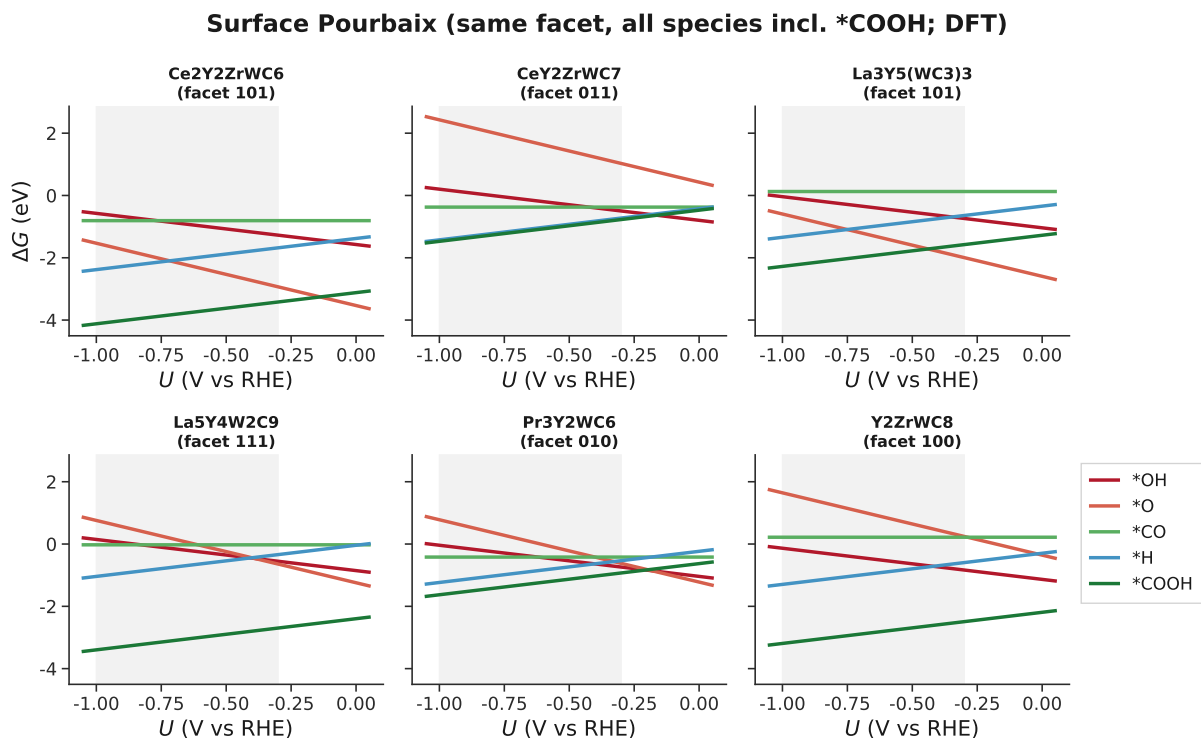


Figure 4.9: Active-facet same-facet full surface Pourbaix of the six candidates (DFT, including *COOH). Potential dependence of each adsorbate’s ΔG ; in the operating window (shaded) the species with the lowest ΔG is the steady state. Across the entire operating potential the reaction intermediate *COOH (green) prevails and the oxidative species (*OH/*O, red) appear only near open circuit, showing that there is no oxidative poisoning during operation.

electrolyte.” The (1) surface *O/*OH Pourbaix and (2) bulk leaching Pourbaix above are *completed* and give a consistent picture: **at the operating cathodic potential the surface is in the *COOH steady state, but at open-circuit/startup conditions the bulk metal passivates (La, Ce, Zr) or leaches (Pr, Y).** Experimentally, this is cross-validated by XPS (surface oxidation state) before and after testing, ICP-MS of the leachate, and the M -O bands of in-situ Raman/SEIRAS (Chapter 5).

4.11 Campaign B — Dual Active-Site Design of Acid-Stable Carbides

Campaign B imposes all four design axioms (A1–A4). The decisive differences from Campaign A are twofold — it internalizes manufacturability (A3) into the search space, and it strengthens selectivity (A2) into an explicit CO/H *dual active-site* criterion. The rationale for excluding rare earths rests on *two independent propositions*, and it is important to keep them separate.

(*Rationale 1 — manufacturability.*) The synthesis route for these carbides metallothermally reduces an oxide precursor with Mg/Ca and then removes the byproduct oxides (MgO/CaO) by *acid leaching*. In this indispensable leaching step the rare earths dissolve entirely (the standard reduction potentials of La/Ce/Pr/Y are very negative and their M^{3+} hydration is strong; Section 4.10), so the candidate pool of Campaign A becomes *already unmanufacturable at the synthesis stage prior to operation*. This is a *synthesis* constraint independent of active-site physics. (*Rationale 2 — non-necessity.*) As shown below, the dual active-site behavior of CO release and H repulsion is explained by the *valence-electron concentration (VEC)* effect of the transition metals (low VEC = group-4-rich = H repulsion) and requires no rare earths at all —

Table 4.4: Bulk aqueous Pourbaix (experimental thermodynamics) verdict for the constituent metals. Stable phase in the CO₂RR operating window (pH 7) and at local alkaline pH 10. Immune = metal, passive = solid oxide/hydroxide, leaching = dissolved ion. *Pure-metal reference (leaching upper bound)*.

Metal	pH 7 (0 ~ -1.1 V RHE)	Local pH 10
La	Passive (La(OH) ₃)	Passive
Ce	Passive (Ce(OH) ₃)	Passive
Pr	Leaching (ion)	Leaching
Y	Leaching (Y³⁺)	Passive
Zr	Passive (Zr(OH) ₄)	Passive
W	Open-circuit leaching / immune under bias	Immune

indeed, the rare earths (group 3, *f*-electrons) lie outside this VEC-controlled regime. That is, the rare earths must be excluded on manufacturing grounds and are *simultaneously* unnecessary for active-site design. Since the two rationales hold independently, the exclusion conclusion survives even if either one is weakened.

Search space and conditioning labels. Campaign B therefore restricts the metal sublattice to *acid-leaching-tolerant* group-4/5/6 transition metals {Ti, Zr, Hf, V, Nb, Ta, Cr, Mo, W} and the nonmetals {C, B, N} (a hard whitelist, `carbide_mattergen/element_policy.py`), and excludes at the source the rare earths (including Sc and Y), the platinum-group metals, and — for *scientific honesty* — the well-known CO₂RR metals {Cu, Ag, Au, Zn}. The latter exclusion is a methodological reason: adding a known CO₂RR metal to a carbide support would obscure the generation model’s contribution to discovery, so to ensure that the activity originates from *new carbide chemistry* these must be excluded. For conditioning labels, instead of a literature-based composition surrogate table, we use *surface labels computed directly with UMA* (the surrogate table is essentially noise, correlating with the UMA labels at $r \approx 0.04$, and so was not adopted). Because UMA surface labeling is expensive, we narrow the Campaign A carbide set to the acid-stable palette and then re-fine-tune CARBIDEMATTERGEN on a high-quality subset re-labeled with UMA (**271 training + 34 validation**; the same adapter scheme and hyperparameters, Section 3.1). Campaign B’s conditioning is thereby grounded directly in surface active-site physics rather than in a compositional surrogate indicator.

HER competition and hydrogen poisoning: the dual objective. A CO₂ → CO active site ($\Delta G_{\text{CO}^*} \approx 0$) often binds H well too, and since HER and CO evolution compete for the site, activity alone is insufficient. In particular, **strong H adsorption is poisoning in which *H covers the surface and eliminates the sites where CO₂ could adsorb**, so the desired condition is a low CO overpotential and *simultaneously* a high HER overpotential — i.e. the surface must *repel* H ($\Delta G_{\text{H}^*} > 0$). We therefore strengthened the candidate criterion into a dual objective:

$$|\Delta G_{\text{CO}^*}| < 0.3 \text{ eV (CO evolution, non-poisoning)} \quad \text{and} \quad \Delta G_{\text{H}^*} > 0 \text{ (H repulsion, site preservation)}.$$

ΔG_{H^*} is defined as the *per-facet* physical-site minimum: the global minimum ΔG_{H^*} is an artifact in which a single dissociative/subsurface H site ($\Delta G_{\text{H}^*} \sim -6$ eV) masks an entire facet as H-repelling, so we take the minimum over the physically adsorbed sites, excluding $|\Delta G_{\text{H}^*}| > 1.5$ eV (dissociation/subsurface), for each facet (`scripts/63_perfacet_dual.py`). This is the requirement that CO evolution and H repulsion hold on the *same facet*.

DFT dual-objective verdict: three Ti-based carbides. Recomputing the 15 species that passed the per-facet dual-objective screen as DFT single-points (DFT//UMA, Γ), **three satisfy**

Table 4.5: Composition/structure and synthesizability indicators of the Campaign B final candidates. VEC is the metal-averaged valence-electron count (group 4/5/6 = 4/5/6). “Acid-stable” denotes survival of acid leaching (A3). E_{hull} (eV/atom) is computed from QE-PBE formation energies against the Materials Project convex hull (GGA/GGA+U): all three are more stable than Campaign A (REE, +0.15 ~ +0.36), and MoTiZr is effectively on the hull (≈ 0).

Candidate	Ideal comp.	Slab (64 atoms)	Structure	Active facet	VEC	Acid-stable	E_{hull}
hec3-MoTiZr	(Mo, Ti, Zr)C	Mo ₁₁ Ti ₁₁ Zr ₁₀ C ₃₂	Rocksalt <i>MC</i>	(100)	4.67	○	-0.02
hec3-CrTiV	(Cr, Ti, V)C	Cr ₁₁ Ti ₁₁ V ₁₀ C ₃₂	Rocksalt <i>MC</i>	(100)	5.00	○	+0.05
hec3-CrTiZr	(Cr, Ti, Zr)C	Cr ₁₁ Ti ₁₁ Zr ₁₀ C ₃₂	Rocksalt <i>MC</i>	(100)	4.67	○	+0.10

Table 4.6: Summary of the DFT active sites and free energies of the Campaign B final candidates. ΔG in eV, U_L in V vs RHE. The active facet is (100) for all; all three simultaneously satisfy CO evolution ($|\Delta G_{\text{CO}^*}| < 0.3$) and H repulsion ($\Delta G_{\text{H}^*} > 0$). The potential-determining step (PDS) is *COOH formation ($\text{CO}_2 + * \rightarrow \text{*COOH}$) for all three candidates.

Candidate	Method	*CO site	ΔG_{CO^*}	ΔG_{COOH^*}	ΔG_{H^*}	U_L	Verdict
hec3-MoTiZr	single-point	Mo ontop (weak)	-0.08	+0.13	+0.40	-0.13	CO-selective
hec3-MoTiZr	relaxed	Mo ontop (released)	+0.18	+0.39	+0.57	-0.40	CO-sel. (conf.)
hec3-CrTiV	single-point	Cr ontop	-0.20	+0.73	+1.01	-0.73	CO-selective
hec3-CrTiZr	single-point	Cr ontop	+0.06	+0.72	+0.71	-0.72	CO-selective

the dual objective, and all are the (100) facet of Ti-based three-metal rocksalt carbides (Table 4.6). The remaining 12 fail by *hydrogen poisoning* with $\Delta G_{\text{H}^*} < 0$ — confirming that *H poisoning is the dominant failure mode*. The Nb/Ta/Mo-rich candidates that the generation model selected on the CO-evolution condition alone released CO but were poisoned by H, whereas the chemistry that repels H was *group-4-rich* (low valence-electron concentration, VEC) such as Ti+Cr/Mo. Indeed, the metal-averaged VEC of the winners is ≈ 4.67 and that of the failures is ≈ 5.5 , showing that **VEC is the physical lever that jointly tunes H repulsion and CO evolution**.

Composition and structure. All three candidates are *equimolar three-metal rocksalt monocarbides* (M, M', M'')C with metal:carbon = 1 : 1 (Table 4.5): hec3-MoTiZr = (Mo, Ti, Zr)C, hec3-CrTiV = (Cr, Ti, V)C, hec3-CrTiZr = (Cr, Ti, Zr)C. The metal counts of the 64-atom SQS (100) slab are 11:11:10, approximating equimolarity (e.g. Mo₁₁Ti₁₁Zr₁₀C₃₂). The metal sublattices are all acid-leaching-stable group-4/5/6 transition metals, so they satisfy the campaign’s design constraint (A3).

Active sites. Looking at the nearest metal to the adsorbate (minimum-image distance) in the relaxed/screened geometries reveals a clear pattern: *CO binds *ontop a group-6 metal* in all three candidates (Cr in CrTiV ~ 1.9 Å, Cr in CrTiZr ~ 2.0 Å, Mo in MoTiZr ~ 2.1 Å). The binding strength differs, however: MoTiZr binds weakly ($\Delta G_{\text{CO}^*} = +0.18 > 0$) and thus *releases* CO, whereas CrTiV binds strongly ($\Delta G_{\text{CO}^*} = -0.20$) and holds it. *H sits ontop/bridge on a metal (e.g. on Mo at 1.76 Å in MoTiZr, on a Zr bridge ~ 2.1 Å in CrTiZr) but is unfavorable in free energy, so its coverage is suppressed. That is, CO binding is tuned by the group-6 metal and H repulsion by the group-4-rich (low-VEC) framework. The (100)-surface CO active sites of the three candidates are shown in Figure 4.11.

Free-energy diagrams: HER suppressed, CO₂RR favored. The CO₂RR ($\text{CO}_2 \rightarrow \text{CO}$) and HER free-energy diagrams of the three candidates are shown together in Fig. 4.12. The CO₂RR pathway is accessible at the applied limiting potential U_L , whereas the HER intermediate *H sits *positive* in free energy for all three candidates ($\Delta G_{\text{H}^*} = +0.57 \sim +1.01$ eV), so *H coverage

is suppressed — i.e. *the same surface suppresses HER while favoring CO₂RR*. CO is, however, the *gateway intermediate*: further reduction beyond CO to multicarbon (C₂₊) products such as ethanol requires calculation of the C–C coupling pathway (Section 6.6) and is therefore only indicated conceptually here (grey dashed) — the full CO₂→ethanol diagram is future work (the earlier CO₂→CO diagram, Fig. 4.7, stops at CO for the same reason).

Full relaxation of the lead candidate: selectivity robust, limiting potential corrected.

For the active (100) facet of the headline candidate hec3-MoTiZr, we fully ionically relaxed the clean slab, *CO, *COOH, and *H (GPU QE, bottom half fixed, Γ). For the same reason as in Section 4.8, the symmetry-free 64-atom-class slab cannot run a $2\times 2\times 1$ k -mesh (16.4 GB) on 16 GB of VRAM, so it was relaxed at Γ , and the cancellation of finite- k error, since ΔG is a same-cell difference, was verified earlier. **The core selectivity claim is robust to relaxation:** $\Delta G_{\text{H}^*} = +0.57$ eV, which is actually *strengthened* relative to the single-point (+0.40), confirming that the H repulsion is not a geometric artifact, and $\Delta G_{\text{CO}^*} = +0.18$ eV, so CO is still weakly bound and desorbs (CO-selective = YES). By contrast, **the limiting potential is corrected:** upon relaxation, *COOH formation ($\Delta G_{\text{COOH}^*} = +0.39$ eV) becomes endothermic and thus the potential-determining step, and $U_L = -0.40$ V (single-point -0.13) worsens. That is, full relaxation *confirms the selectivity while correcting the apparent activity*, revealing that the single-point DFT//UMA underestimated the *COOH barrier — demonstrating exactly the value of relaxation validation. The relaxed final structure (Figure 4.13) shows the dual selectivity directly through geometry: after relaxation *CO remains weakly bound *ontop* Mo (~ 2.1 Å) but, since $\Delta G_{\text{CO}^*} = +0.18 > 0$, is thermodynamically *released* rather than poisoning the surface, while *H adsorbs on the surface (~ 1.7 Å) but is unfavorable in free energy ($\Delta G_{\text{H}^*} = +0.57$ eV), keeping its coverage low.

Retargeting the generator: low-VEC conditioning drives discovery. The three winners above came from the high-entropy SQS ladder (Section 4.9) and were not direct products of the generation model — because the initial CO conditioning was biased toward Nb/Ta/Mo-rich (H-poisoned) compositions. We therefore retargeted the conditioning to *low VEC* (4.5) and regenerated 2048 structures, whereupon **the generation model itself produced dual-objective winners** (three species by the UMA per-facet screen: HfZrTa₂Nb-C (110) $\Delta G_{\text{CO}^*} = -0.06/\Delta G_{\text{H}^*} = +0.22$, HfTa₂Nb-C (011) $-0.10/+0.21$, HfZr₃TaNb-C (011) $-0.13/+0.18$; all group-4/5-rich, low VEC). Their repulsion margin is smaller than that of the SQS winners ($\Delta G_{\text{H}^*} = +0.4$ to $+1.0$), but the key point is that the *H-repelling, CO-evolving surfaces that previously came only from the SQS ladder are now proposed directly by the generation model*, confirming the causal effectiveness of the VEC lever.

Honest caveats. (i) The three confirmed winners are explicit SQS solid solutions, not generation products (a generation-product winner was separately confirmed by the retargeting above). (ii) hec3-CrTiV/CrTiZr are single-points (DFT//UMA), though their H-repulsion margin is large ($\Delta G_{\text{H}^*} = +1.01/+0.71$), making them less relaxation-sensitive than the lead candidate. (iii) The lead-candidate relaxation is a Γ -level local relaxation (loose force convergence 0.26 eV/Å), better than a single-point but not k -converged — the ~ 0.5 eV geometric shift observed during relaxation itself corroborates this limitation of the single-point. (iv) It is an equimolar SQS idealization, and an *ordered* small surrogate cell cannot reproduce this dual behavior (upon ordering, the CO-evolution facet and the H-repulsion facet separate) — i.e. the dual activity is a property *specific to disorder/composition*. (v) $\Delta G_{\text{CO}^*} \approx 0$ means a *CO source*, not C–C coupling, and C₂₊ targets such as ethanol require a separate coupler paired with these CO-releasing carbides.

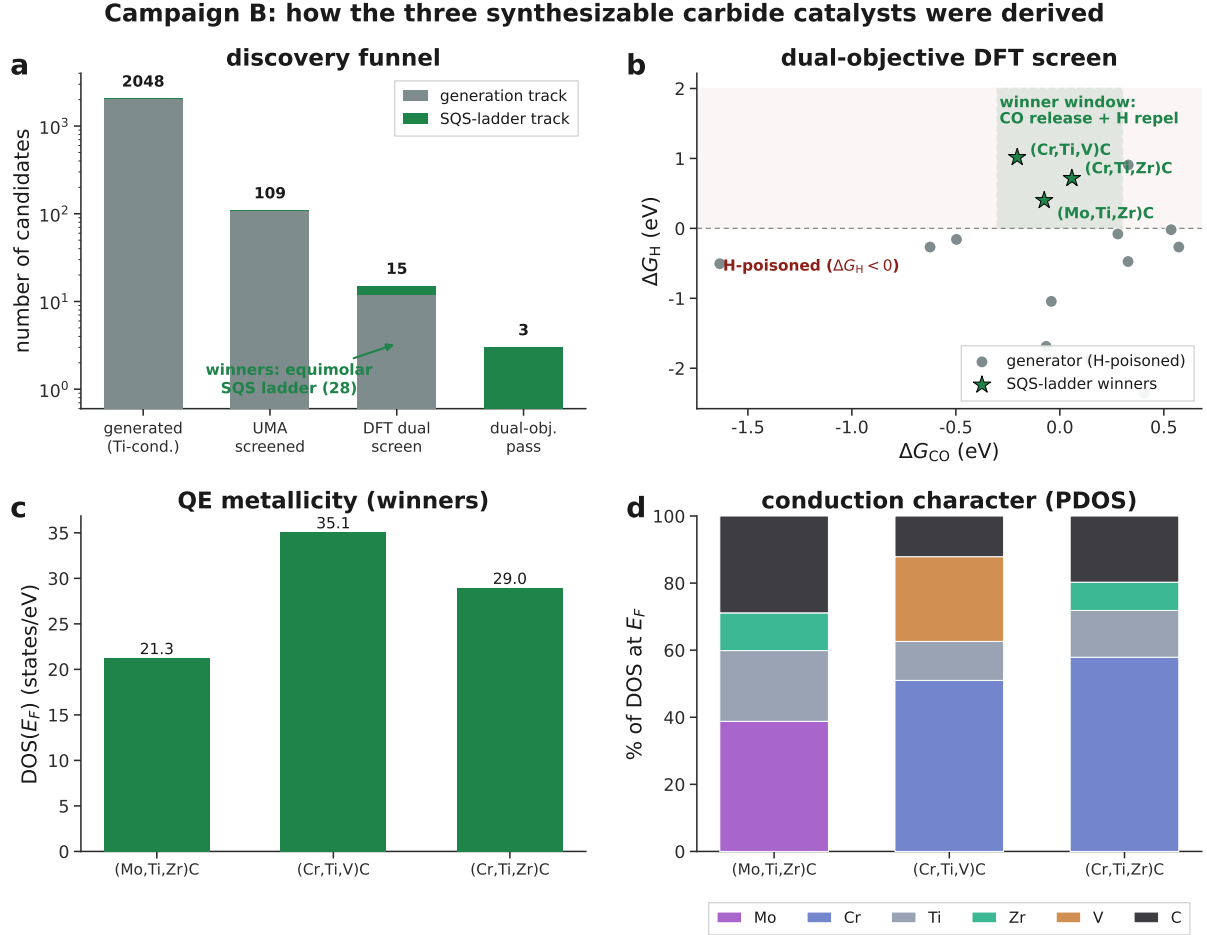


Figure 4.10: End-to-end derivation of the Campaign-B candidates — the carbide analog of Figure 4.1 (Campaign A). (a) Discovery funnel: 2048 Ti-conditioned generated structures $\rightarrow$ UMA screen $\rightarrow$ 15 DFT dual-objective-screened $\rightarrow$ 3 that pass. The *generation track* (grey) sends 12 candidates to DFT that are *all* lost to H-poisoning (0 pass), whereas the 3 winners come from the *equimolar SQS ladder* (green) — i.e. the generator under-samples the H-repelling Ti-rich chemistry, so the winners were secured by systematic SQS construction. (b) DFT dual-objective screen: ΔG_{CO}^* vs ΔG_H . Only the three Ti-based carbides (stars) fall in the winner window (CO release $|\Delta G_{CO}^*| < 0.3$ and H repulsion $\Delta G_H > 0$); the 12 generator candidates are poisoned ($\Delta G_H < 0$). (c) QE bulk metallicity DOS(E_F) of the three winners. (d) Element-projected conduction character (per-element share of the DOS at E_F). Panels (c,d) are at the same computational level as Figure 4.1(c,d) (QE bulk SCF $\rightarrow$ dos $\rightarrow$ projwfc).

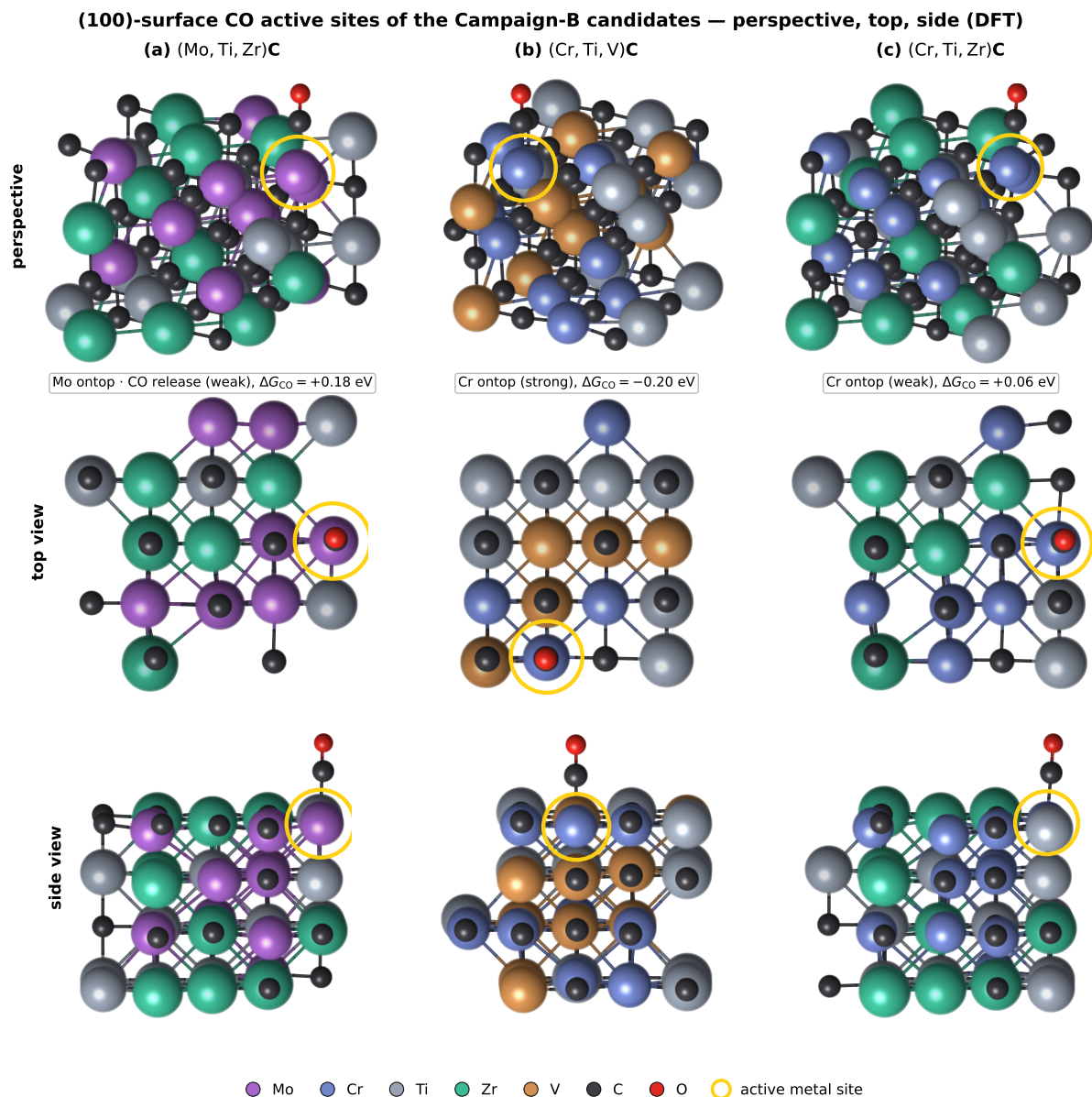


Figure 4.11: (100)-surface CO active sites of the three Campaign B final candidates (DFT relaxed/screened geometries). **Columns** are the three candidates; **rows** are the perspective (3D), top view (looking straight down), and side view (from the front). In each panel the yellow ring marks the nearest (active) metal site of the adsorbed CO. The top view places the CO oxygen (red) directly over the active metal, and the side view shows the CO standing vertically above the surface in an *ontop* geometry, making the active location unambiguous. (a) (Mo, Ti, Zr)C binds CO weakly *ontop* Mo (~ 2.1 Å) but, since $\Delta G_{CO^*} = +0.18 > 0$, *releases* it (relaxed). (b) (Cr, Ti, V)C and (c) (Cr, Ti, Zr)C bind CO at the *Cr ontop* site (~ 1.9 – 2.0 Å). For all three candidates, CO binding/release is tuned by the group-6 (Cr/Mo) site. See the legend for atom colors; the active facet is (100) for all.

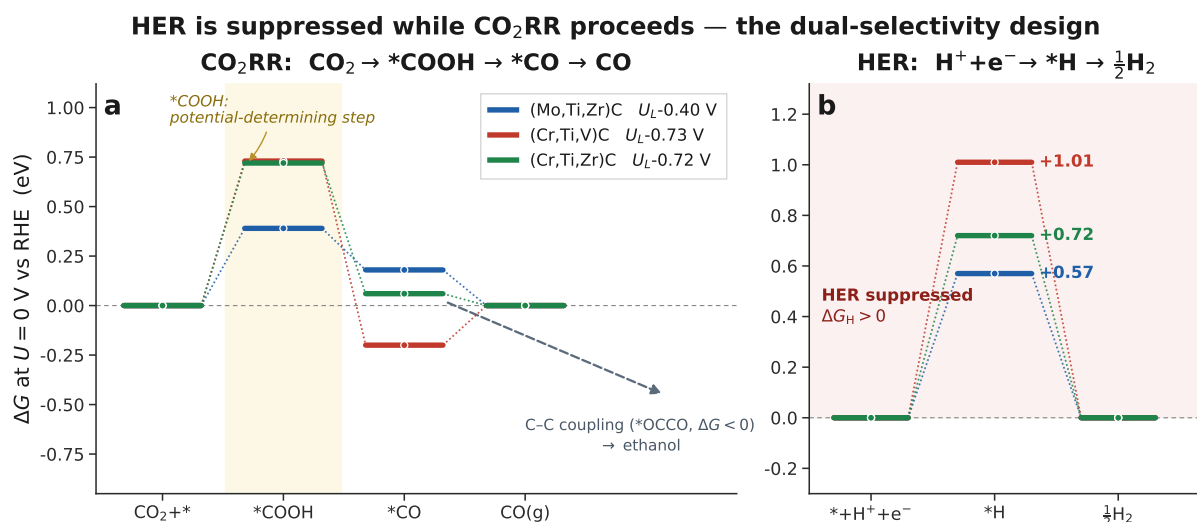


Figure 4.12: Free-energy diagrams of the three Campaign-B final candidates ($U = 0$ vs RHE). (a) CO₂RR: CO₂ → *COOH → *CO → CO; the potential-determining step is *COOH formation for all three. Route beyond *CO to ethanol (C₂₊): the first C–C-coupling intermediate (*OCCO, C–C ≈ 1.5 Å, $\Delta G < 0$) was computed and shows the C2 (ethanol) route is favored (C1 *CHO is endergonic; Section 6.6); the grey dashed arrow denotes that the *full* downstream pathway (transition states / complete route) is still conceptual. (b) HER: H⁺ + e⁻ → *H → $\frac{1}{2}$ H₂; the *H level is positive ($\Delta G_H^* > 0$) for all three, so HER is suppressed (shaded region). The top of the y -axis is matched across panels so the CO₂RR rate-limiting level (*COOH) and the HER *H height can be compared directly. hec3-MoTiZr is relaxed; the other two are single-point.

hec3-MoTiZr (100): DFT-relaxed final structure — CO₂RR-selective ($U_L = -0.40$ V, H-repelling)

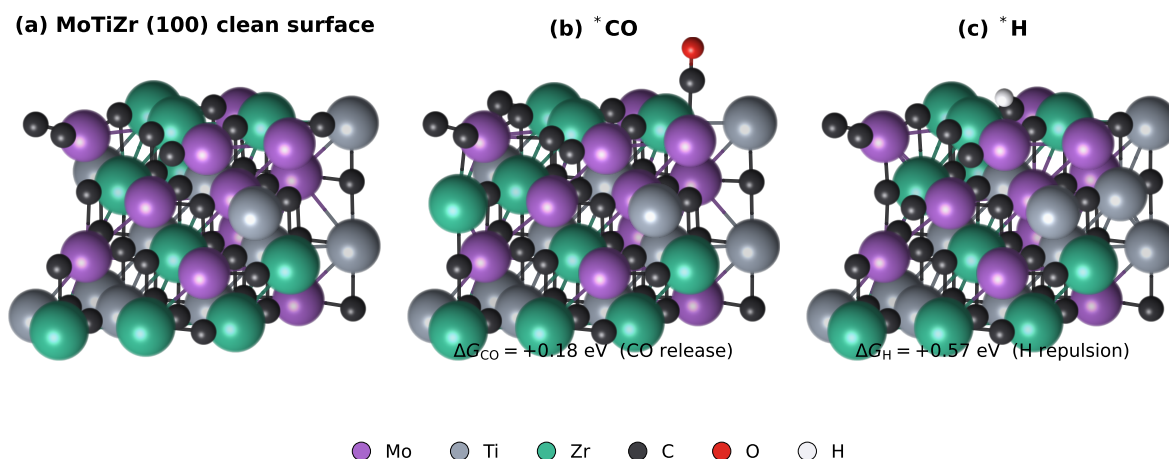


Figure 4.13: DFT fully relaxed final structure of the headline candidate **hec3-MoTiZr (100)** (SQS 64-atom slab, bottom half fixed, Γ). (a) clean surface, (b) *CO, (c) *H. The dual objective is revealed geometrically: after relaxation *CO stays weakly bound *ontop* Mo (~ 2.1 Å) but is *released* thermodynamically ($\Delta G_{CO^*} = +0.18$ eV, weak binding), while *H adsorbs on the surface but is unfavorable in free energy ($\Delta G_H^* = +0.57$ eV, H repulsion), suppressing its coverage. That is, the surface leaves the CO₂RR intermediate site vacant while suppressing HER ($U_L = -0.40$ V vs RHE, CO₂RR-selective). Atom colors: Mo purple, Ti gray, Zr green, C dark gray, O red, H white. Coordinate files are `results_cif/MoTiZr_*.cif`.

Chapter 5

Planned Experimental Validation

Experimental validation targets the *manufacturable* Campaign-B final candidates (hec3-MoTiZr as the representative, together with CrTiV/CrTiZr) — the rare-earth candidates of Campaign A dissolve during acid leaching and thus cannot become test subjects in the first place, and it is precisely that constraint which defined the acid-stable transition-metal palette of Campaign B (Section 4.11). The pipeline yields concrete, falsifiable predictions: these candidates should (i) be synthesizable as single-phase metallic carbides and (ii) electroreduce carbon dioxide to carbon monoxide with a selectivity that outperforms HER. This chapter specifies the experimental program designed to test this. Results will be reported upon completion of the campaign.

5.1 Materials Synthesis

The Campaign-B final candidates (Ti-based transition-metal carbides) are synthesized by high-temperature carburization of ball-milled metal/oxide precursors (or arc melting followed by carburization), targeting the near-equimolar metal sublattice predicted by the SQS. Even when the metallothermic-reduction–acid-leaching route is used, this composition withstands the leaching step (a consequence of designing with an acid-stable palette). Phase purity and the single-phase rocksalt/hexagonal structure are verified by powder X-ray diffraction (XRD), the homogeneity of the metal distribution by SEM/EDS and STEM-EDS mapping, and the surface composition/oxidation state by X-ray photoelectron spectroscopy (XPS). **[Synthesis and structural analysis – to be updated upon experimental completion]**

5.2 Electrochemical CO₂RR Testing

The catalyst ink is drop-cast onto a gas-diffusion or glassy-carbon electrode and tested in a gas-tight H-cell (and, for high current densities, an MEA/flow cell) in CO₂-saturated aqueous bicarbonate electrolyte. Partial current densities are quantified by linear-sweep voltammetry and potentiostatic measurements over a series of applied potentials, and activities normalized by geometric and electrochemically active surface area are reported. **[Polarization curves and potential-resolved Faradaic efficiencies – to be updated upon experimental completion]**

5.3 Product Analysis

- **Gaseous products (CO, H₂, residual CO₂).** The cell headspace/outlet is sampled online by gas chromatography (TCD + FID with a methanizer) to quantify CO and H₂, from which the CO Faradaic efficiency and CO/H₂ selectivity are computed, while CO₂ conversion is tracked by an inlet/outlet mass balance. Online differential electrochemical

mass spectrometry (DEMS) provides the potential-resolved onset of CO/H₂. **[GC/DEMS gaseous-product quantification – to be updated upon experimental completion]**

- **Liquid products.** The catholyte is analyzed by quantitative ¹H nuclear magnetic resonance (NMR) spectroscopy with water suppression and an internal standard (e.g., DMSO/DSS). *Ethanol was detected by ¹H NMR in preliminary electrolysis* (along with other liquid products such as formate), showing that multi-carbon (C₂₊) production actually occurs *beyond* the two-electron CO pathway that the design targeted. This observation is interpreted as a *multi-site tandem* in which the site diversity of the high-entropy surface mediates both CO production and C–C coupling within a single material (Section 6.6). Quantification of the concentration and Faradaic efficiency of each liquid product, together with the ethanol-versus-CO partitioning, is reported in **[¹H NMR liquid-product quantification (including ethanol FE) – to be updated upon experimental completion]**
- **Adsorbed intermediates.** In-situ attenuated-total-reflection surface-enhanced infrared absorption spectroscopy (ATR-SEIRAS / FT-IR) detects the potential-dependent *COOH and linear/bridge-bonded *CO bands, providing a direct mechanistic link to the computed ΔG_{COOH^*} and ΔG_{CO^*} descriptors. **[In-situ FT-IR of the *COOH/*CO intermediates – to be updated upon experimental completion]**

5.4 Stability and Post-Mortem Analysis

Durability is assessed by long-duration electrolysis, and XRD, XPS, and microscopy before and after testing probe surface oxidation, leaching of the rare-earth constituents, and the integrity of the carbide network. **[Durability and post-mortem characterization – to be updated upon experimental completion]**

5.5 Validation Logic

The experiments are designed to test the computational predictions item by item: metallic conductivity (four-point probe / electrochemical impedance) against the DFT DOS, CO selectivity (GC/NMR) against the computed ΔG_{CO^*} and the $\Delta G_{\text{COOH}^*} < \Delta G_{\text{H}^*}$ selectivity criterion, and the mechanistic intermediates (FT-IR) against the descriptor ladder. The prediction-measurement comparison table is reported upon completion (Table 5.1).

Table 5.1: Planned comparison of the computational predictions and the experimental observables.

Prediction (this work)	Experimental means of validation	Status
Single-phase HEC, equimolar metals	XRD, STEM-EDS	[Result – to be updated upon experimental completion]
Metallic (finite DOS at E_F)	Four-point / EIS	[Result – to be updated upon experimental completion]
CO selectivity, $\Delta G_{\text{COOH}^*} < \Delta G_{\text{H}^*}$	GC, ^1H NMR, FE	[Result – to be updated upon experimental completion]
*COOH→*CO pathway	In-situ FT-IR	[Result – to be updated upon experimental completion]
CO ₂ conversion	Inlet/outlet GC balance	[Result – to be updated upon experimental completion]

Chapter 6

Discussion

6.1 Why This Pipeline, and Its Advantages

The central methodological claim of this thesis is that *the economic constraint belongs in the generative stage*. By fine-tuning the diffusion model on expensive-metal-fraction labels and guiding toward zero, the inexpensive subspace is sampled *constructively*, in contrast to the way one would filter a sparse set of unconstrained candidates to discover it. As a result, the funnel surveys $\sim 10^3$ candidates while using DFT on only six structures, and each stage (composition surrogate $\rightarrow$ MLIP $\rightarrow$ DFT) adds accuracy exactly where the previous stage has already eliminated most of the search space. This is a generalizable framework: swapping the conditioning descriptors re-aims the same machine at a different reaction.

6.2 Interpretation of the Candidates

Below is an interpretation of the six candidates of *Campaign A* (the activity-first baseline), showing that the pipeline captures a valid activity/conductivity descriptor region. These candidates, however, are eliminated at the manufacturability gate (Section 4.10); therefore the *manufacturable* final catalyst is the Ti-based carbide of Campaign B (represented by hec3-MoTiZr, Section 4.11). All six candidates combine inexpensive early transition metals (Y, Zr, W) with light rare earths (La, Ce, Pr) on a carbide metal sublattice. The conduction character elucidated by DFT (Figure 4.5) shows a continuum from carbon-2p-network metals to metal-*d* metals, reflecting the “platinum-like” picture in which carbon reconstructs the metal *d* states [12]. The near-thermoneutral ΔG_{CO^*} and the $\Delta G_{\text{COOH}^*} < \Delta G_{\text{H}^*}$ selectivity place these materials in the same descriptor region that makes Au/Ag/Zn CO-selective [9], but realize it with earth-abundant constituent elements — which is precisely the design goal. The success of high-entropy alloys as experimental CO₂/CO reduction catalysts [11, 10] and the success of Mo₂C for selective CO₂-to-CO [13] support the external validity of the carbide + HEC route.

6.3 Surface Oxidation / OH Poisoning: Tentative Conclusions and Compositional Prescription

Because of the large oxygen affinity of the rare earths, “does it not oxidize in aqueous solution and cover the active sites” is a central concern for this materials family. We organize the results of examining this via a same-facet full-surface Pourbaix analysis (*CO/*COOH/*H/*OH/*O, Section 4.10) as tentative conclusions.

Under operating conditions, it is not oxidative poisoning. When the steady state is judged on the active facet of each candidate, including the reaction intermediate *COOH, at

Table 6.1: La content and oxidation tendency (*OH binding) by composition. The deeper the worst-facet ΔG_{*OH} , the greater the open-circuit oxidation risk. The active-facet values show that binding is much weaker (less oxidative) on the actual reaction facet. Units eV.

Candidate	La% of metals	Worst-facet ΔG_{*OH}	Active-facet ΔG_{*OH}
$\text{La}_5\text{Y}_4\text{W}_2\text{C}_9$	45	-6.67	-0.85
$\text{La}_3\text{Y}_5(\text{WC}_3)_3$	27	-2.41	-1.04
$\text{Ce}_2\text{Y}_2\text{ZrWC}_6$	0	-3.85	-1.57
$\text{CeY}_2\text{ZrWC}_7$	0	-3.39	-0.80
$\text{Pr}_3\text{Y}_2\text{WC}_6$	0	-3.09	-1.04
Y_2ZrWC_8	0	-3.30	-1.13

the operating potentials ($-0.3 \sim -1.0$ V vs RHE) all six candidates are dominated by *COOH (the operating state) rather than by *O/*OH. Oxidative coverage is confined to the transient state near open circuit (~ 0 V). In other words, the stage of risk is not *normal operation* but *startup/open-circuit and exposure to air and moisture*.

The problematic compositions: La-rich compositions. The oxidation *tendency* (worst-facet *OH binding strength) is strongly coupled to the La content of the composition (Table 6.1). $\text{La}_5\text{Y}_4\text{W}_2\text{C}_9$ (La 45% of the metal sublattice) is overwhelmingly deep at $\Delta G_{*OH} = -6.7$ eV, carrying the greatest risk of open-circuit oxidation/leaching, and was also the only composition for which the GPU SCF failed to converge computationally. $\text{La}_3\text{Y}_5(\text{WC}_3)_3$ requires the largest cathodic bias (-0.6 V) to reach the operating state on its active facet. By contrast, the La-free Ce/Pr/Y-based compositions ($\text{CeY}_2\text{ZrWC}_7$, $\text{Pr}_3\text{Y}_2\text{WC}_6$, etc.) have moderate *OH binding (below the -3 eV range) and switch to the operating state at gentler potentials. The standard reduction potential of La (-2.38 V) is likewise more negative than those of Ce/Pr, consistent with the picture that *La is the principal risk element for oxidation/leaching*.

Resolution (prescription). (1) *Compositional design*: guide the generator to suppress the La content (broadly, the fraction of oxidation-prone REEs). Reusing the core technique of this pipeline exactly — just as economics (the fraction of expensive metals) was conditioned — we **add the surface *OH binding (or oxide-film formation) descriptor as a tenth conditioning property** to constructively sample the low-oxidizability subspace. *Jointly* conditioning on activity ($\Delta G_{CO} \approx 0$) and oxidation resistance (weak *OH) prioritizes $\text{CeY}_2\text{ZrWC}_7$ -type candidates (low overpotential + low oxidizability). (2) *Operating protocol*: since *COOH dominates at the operating potential, minimize open-circuit dwell and rapidly apply cathodic bias, and carry out synthesis/transfer under exclusion of air and moisture. (3) *Kinetic protection*: because the bulk is metastable (Section 4.7, $+0.15 \sim 0.36$ eV/atom above the hull) and the leaching driving force of La is large, a strategy of suppressing startup oxidation with a surface passivation film or protective layer may be effective. These prescriptions are directly verified by the XPS (surface oxidation state), ICP-MS (leaching), and in-situ FT-IR (*COOH/*CO bands) of Chapter 5.

Provisionality. The above conclusions are tentative because (i) the worst facet is a worst-case bound and the actual effective coverage is unquantified, (ii) the MLIP-DFT agreement is low for weakly bound *OH ($r=0.74$), with DFT giving weaker binding, and (iii) the bulk Pourbaix numerical diagram is incomplete; experiment is the final arbiter.

6.4 Assessing the Synthesizability of the Campaign-B Candidates

The three final candidates (Table 4.5) are all equimolar three-metal rocksalt monocarbides (M, M', M'')C. Their practical synthesizability is assessed by the following factors.

Favorable factors. (i) *Acid-leaching stability* ($A3$, *satisfied by design*). Because the metal sublattice is entirely group-4/5/6 transition metals (Ti, Zr, V, Cr, Mo), it does not dissolve — unlike the rare earths — during the leaching step of the metallothermic-reduction–acid-leaching route; this constraint defined the palette in the first place (Section 4.11). (ii) *Structural precedent*. Equimolar multi-metal rocksalt carbides are a synthesized materials family, and refractory five-metal rocksalt high-entropy carbides have been discovered and synthesized via entropy descriptors [14]; three-metal compositions are more accessible than those. (iii) *Stable binary skeleton*. Among the constituent binaries, TiC, ZrC, and VC are thermodynamically stable rocksalt phases and thus stabilize the skeleton.

Convex-hull stability (computed). Using the *same* recipe as Campaign A (QE-PBE formation energy + Materials Project GGA/GGA+U convex hull), we computed the above-hull energy E_{hull} of the three candidates (Table 4.5): **hec3-MoTiZr** ≈ -0.02 (**effectively on the hull, stable**), **hec3-CrTiV** $+0.05$, **hec3-CrTiZr** $+0.10$ eV/atom. All three are *more stable* than the REE carbides of Campaign A ($+0.15 \sim +0.36$, Section 4.7), and MoTiZr sits on the hull. The competing decomposition phases predicted by MP are stable binary carbides (TiC, ZrC, Mo₂C, Cr₃C₂, V₆C₅), with a metastability margin of only $\leq +0.10$ eV/atom — well within reach of configurational-entropy stabilization. In short, the Campaign-B candidates are “manufacturable (acid-stable) *and* thermodynamically on-hull to low-metastable,” qualitatively different from the unmanufacturable REE candidates of Campaign A.

Residual risk factors. (i) Since the rocksalt phases of CrC and δ -MoC are metastable (or high-pressure phases) in the pure state, solid solutions containing them depend on *configurational-entropy stabilization*, but the $E_{\text{hull}}(\leq +0.10)$ above quantitatively confirms that this margin is small. (ii) A methodological uncertainty of ~ 0.1 eV/atom arises from the QE(SSSP)-vs-MP(VASP) reference difference (largely cancelled by the elemental references but with a residual), Cr is treated non-magnetically, and MoTiZr’s negative E_{hull} is interpreted as *on the hull* within this error. (iii) Since the SQS idealizes equimolar, fully disordered occupancy, the compositional deviation, partial ordering, and possible phase separation of the actual synthesized material must be confirmed by XRD/EDS (Chapter 5).

Summary. The acid-leaching constraint is already satisfied by design and the structural precedent is positive, but confirmation of the *thermodynamic* synthesizability rests on the E_{hull} calculation and an actual phase-purity test. In short, the three candidates are “metastable high-entropy carbides with a secured (acid-stable) manufacturing route,” qualitatively distinct from the “unmanufacturable rare-earth candidates” of Campaign A.

6.5 Capabilities and Limits of the Generative Model: High-Entropy Extrapolation and the SQS Ladder

The three confirmed winners are explicit SQS solid solutions, not direct products of CARBIDE-MATTERGEN (Section 4.11). This is not a defect to be hidden but a point that honestly defines the *role and limits of the generative model* within the generative pipeline, and its causes and responses are synthesized in Figure 6.1.

Proximate cause — training distribution. The training set of CARBIDEMATTERGEN (the 5,275 pure carbides of `alex_mp_20`) has a metal-species-count distribution that forms a *cliff at three metals* (Figure 6.1a): ≥ 4 metals constitute only 0.04% of the total (two four-metal, zero five-metal), and the mixing entropy tops out at only $\sim 1.1 R (= \ln 3)$. Thus the conditioning target $S_{\text{mix}} = 1.6 R$ (corresponding to equimolar five metals) is an extrapolation *outside* the training distribution, and at both guidance strengths the four-metal yield remains at $\sim 1\%$ and the model merely piles on more Mo. That is, the generative model cannot fabricate high-entropy compositions it has never learned.

Root cause — sublattice structure and thermodynamics. This data skew itself is rooted in physics (Figure 6.1b). Rocksalt monocarbides have *only one* metal sublattice, so different metals must share the same site; when the differences in atomic size and electronic structure are large (especially the metastable rocksalt phases of the group-6 Cr/Mo/W), the solid solution becomes unstable. As a result, high-entropy carbides are metastable at 0 K and hence rare in natural and computational databases, which feeds back into the training-set scarcity. In short, “limited training data” and “structural instability” are not separate but *two faces of the same cause*.

Contrast — why oxides reach five components. In the oxide version of the same pipeline (OxideMatterGen), five-component inert anodes were obtained by generation. The difference lies not in the algorithm but in the chemistry/structure: oxides, like perovskites (ABO_3), have *two cation sublattices*, *A* and *B*, dispersing different elements over multiple sites, and their strong ionic bonding permits diverse oxidation states and ionic radii, while high-entropy oxides (HEOs) are experimentally established. Indeed, that oxide training set *does contain* ≥ 4 cations at 2.1% (including five- and six-metal, on the order of ~ 600 in absolute number), so the generative model could learn and reproduce that distribution (Figure 6.1a, blue). The single-sublattice rocksalt skeleton of the carbides does not afford this latitude.

Role of the SQS ladder and the honest boundary. The SQS ladder is therefore not a separate task independent of the generative model, but a complementary tool that extends the *palette and operating chemistry* defined by the generative model (the group-6 + group-4/5 combination) to high entropy — circumventing by explicit SQS construction the extrapolation that generation cannot perform (Figure 6.1c). In short, the contribution of the generative model in this pipeline is not the direct “invention” of the final structure but (i) narrowing a vast composition space to the four design goals and (ii) revealing the operating-chemistry pattern and the physical lever (VEC). And (iii) it was confirmed that, when re-aimed at low VEC, the generative model *directly* yields dual winners within the three-metal range (Section 4.11). Making explicit the limits of generative extrapolation and their structural origin is the honest boundary of this methodology, and to enable high-entropy generation directly for carbides one must either broaden the training distribution to include multi-sublattice skeletons or separate entropy into an explicit construction step.

6.6 Multi-Carbon Products and the Multi-Site Tandem Hypothesis

The computational design of this work targeted only $\text{CO}_2 \rightarrow \text{CO}$ (two electrons) and HER suppression, yet the ^1H NMR of the preliminary electrolysis showed *ethanol* production (Chapter 5; quantification in progress). Ethanol is a 12-electron, multi-carbon (C_{2+}) product that requires *C–C coupling*, so this is an observation beyond the design goal and requires honest interpretation.

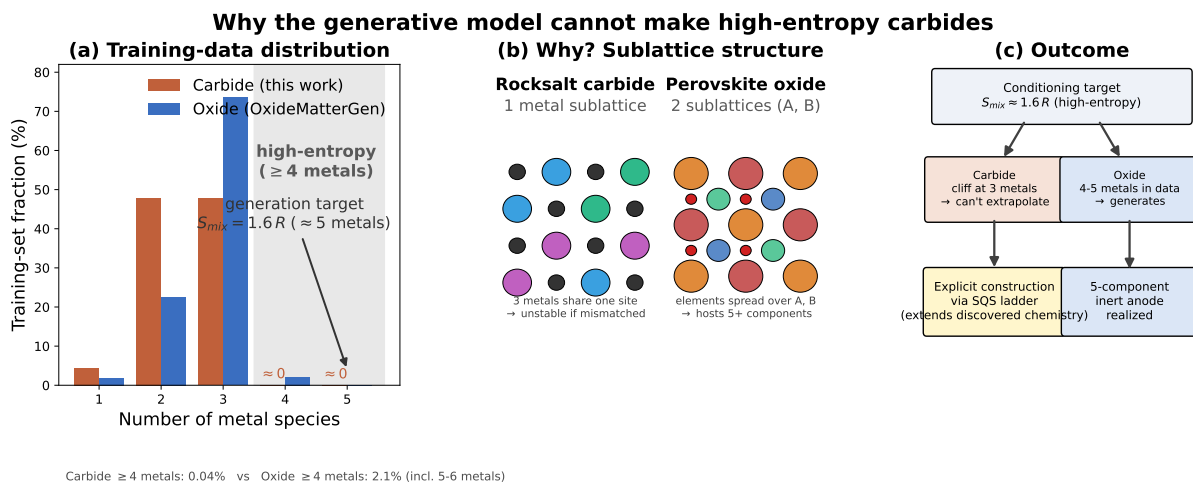


Figure 6.1: Why the generative model cannot make high-entropy carbides, and the role of the SQS ladder. (a) Training-set metal-species-count distribution (measured): the carbides (orange) form a cliff at three metals (≥ 4 metals 0.04%), whereas the oxides (blue, OxideMatterGen) do contain 4–6 metals (2.1%); the conditioning target $S_{\text{mix}} = 1.6R$ is an extrapolation outside the carbide distribution. (b) Root cause: rocksalt carbides have one metal sublattice (multiple metals compete for one site $\rightarrow$ unstable), whereas perovskite oxides have two sublattices, A and B (elements accommodated dispersively $\rightarrow$ 5+ components stable). (c) Consequence: for carbides, generative extrapolation fails $\rightarrow$ explicit construction via the SQS ladder; for oxides, five components are realized by generation.

Tension from the single-site view. At first glance this seems to conflict with the design: the optimized active site has $\Delta G_{\text{CO}^*} \approx 0$ (weak binding / release), which by a *single-site* criterion is actually unfavorable to C–C coupling — because CO leaves the surface before coupling can occur. That is, the very site we optimized is a CO *source*, not a coupling site.

Multi-site tandem hypothesis. This tension is resolved by the essence of the *high-entropy* surface — site diversity. The screening ranked each composition by its *optimal* site ($\Delta G_{\text{CO}^*} \approx 0$), but an equimolar multi-metal rocksalt surface has a broad ΔG_{CO^*} distribution, and sites that bind CO more strongly also exist. These strong-binding sites retain CO and let it couple with an adjacent CO to proceed to C_{2+} , while the weak-binding sites handle the $\text{CO}_2 \rightarrow \text{CO}$ supply. Thus *a single material can act as a tandem catalyst possessing both CO-production sites and C–C coupling sites*, and the observed ethanol is interpreted as this emergent multi-site effect. Furthermore, H repulsion (A_2 , $\Delta G_{\text{H}^*} > 0$) is favorable not only to CO but to *all* CO_2RR products — by suppressing both current loss to HER and *H poisoning, it also contributes to C_{2+} selectivity.

Branching verdict: which downstream route is favored (computed). To partially test this hypothesis, we computed with UMA the first intermediate of the three fates of *CO — CO release, C1 hydrogenation (*CHO), and C2 coupling (*OCCO) — on the (100) surface of all three candidates (adding $\text{*CHO}/\text{*OCCO}$ to the pipeline). The C–C-coupled *OCCO dissociates into two CO at most sites, but a physically coupled site ($\text{C–C} \approx 1.5 \text{ \AA}$) survives on all three candidates and its adsorption free energy is *negative* ($\Delta G_{\text{*OCCO}} = -0.56 \sim -1.17 \text{ eV}$ at the coupled site), showing that **the surface stabilizes CO–CO coupling**. In contrast, C1 hydrogenation (*CHO) is generally *endergonic* ($\sim +0.8 \text{ eV}$). Thus the calculation indicates that the **C2 (ethanol) route is favored over C1 (methanol/methane)**, in qualitative agreement with the experimental observation of ethanol (indicated conceptually in Fig. 4.12a).

Limitations and future work. The branching verdict above is a *falsifiable* partial check, but with several caveats: (i) only the *first* *OCCO coupling intermediate was computed — the *full* ethanol pathway (subsequent proton–electron transfers) and the coupling *transition states / barriers* are not computed, and since most sites dissociate (coupling survives only at specific sites), C–C coupling is a UMA-level (out-of-training-distribution) *qualitative* signal that requires DFT verification (the *OCCO/*CHO zero-point corrections are also approximate); (ii) since ethanol arises from the *site distribution* rather than the optimal site, it is not directly linked to the activity ranking; (iii) the ethanol-versus-CO partitioning (Faradaic efficiency) is unquantified. Future work is to compute from first principles the CO–CO coupling energetics across the site distribution and to *add a coupling descriptor as a conditioning property* so as to actively design C₂₊ selectivity. In short, this thesis designed and validated the *necessary condition* of CO production and HER suppression, and ethanol is an encouraging experimental result shown by the high-entropy surface on top of that, indicating an extension pathway toward C₂₊.

6.7 Limitations and Honest Caveats

Several caveats must be stated. (i) The light rare earths (La, Ce, Pr, Y) are economically “inexpensive” by the chosen descriptor, but rare-earth carbides are unusual CO₂RR catalysts, oxophilic, and may be sensitive to air/moisture. Experimental robustness is an open problem that the planned stability tests address directly. (ii) The MLIP surface screening was calibrated to offset baseline error, but it evaluates rare-earth carbide surfaces that lie mostly outside the learned distribution. In particular, the strongly bound *COOH energy must be read qualitatively, and the activity ranking relies on the more robust ΔG_{CO^*} descriptor. (iii) The composition surrogate used in Stages A–B is a literature-based estimate for coarse filtering, not a quantitative prediction. (iv) DFT treats the rare-earth 4*f* electrons as a fixed trivalent core. This is appropriate for these metallic carbides, but properties sensitive to the *f*-states require a 4*f*-valence or DFT+*U* treatment. (v) The current DFT verifies *bulk* stability and metallicity. Explicit disordered *surface* free energies in DFT (on the SQS supercell of Stage A') are the natural next computational step beyond the MLIP-level surface screening reported here. These limitations motivate rather than weaken the experimental campaign of Chapter 5, and experiment is the arbiter of the predictions.

6.8 Outlook

Beyond confirming the Ti-based candidates of Campaign B, the contrast between the two campaigns demonstrated the value of *placing the constraint in the generative stage*: when the generator was re-aimed at low VEC in Campaign B, the model itself yielded H-repelling candidates, suggesting that discovery can come from conditioning rather than a post-processing screen. The pipeline supports an active-learning loop in which experimental and higher-level DFT data recalibrate the composition surrogate and the MLIP to refine successive generations. The same conditional generator can be re-aimed at CO₂-to-formate or multi-carbon products by changing the conditioning descriptors.

Chapter 7

Conclusion

This thesis internalized the four design axioms of a superior CO₂RR carbide catalyst (activity, selectivity, manufacturability, economics) in a property-conditioned diffusion model and, *driving a single pipeline with two constraint sets*, validated them through a funnel of machine-learning potentials and first-principles DFT. *Campaign A* (the activity-first baseline) derived six platinum-group-free high-entropy carbides from 1024 generated structures and characterized them by surface free energy, *d*-band, convex hull, and same-facet surface Pourbaix (at the operating potential the steady state is the reaction intermediate *COOH, not oxidation). This candidate group, however, is eliminated at the *manufacturability gate* — the rare earths dissolve during the essential leaching step of metallocathodic-reduction–*acid-leaching* synthesis and are thus unmanufacturable. This is not a failure but a controlled contrast, showing that *unconstrained activity search converges to an unmanufacturable chemistry* and demonstrating the need to promote manufacturability to a generative constraint.

Campaign B imposes all four axioms: it restricts the search space to acid-leaching-stable group-4/5/6 transition metals (excluding, along with rare earths and platinum-group metals, the known CO₂RR metals Cu/Ag/Au/Zn as well, to secure the discovery contribution of the generative model), conditions on UMA-computed surface labels, and explicitly addresses HER competition — because strong H adsorption is a *poisoning* that covers the CO₂ sites — applying a *dual active-site* criterion that simultaneously demands CO evolution ($|\Delta G_{\text{CO}^*}| < 0.3$) and *H repulsion* ($\Delta G_{\text{H}^*} > 0$) on the same facet. Rare earths are excluded for manufacturing and are also *unnecessary* for the active-site design, because the dual behavior is a valence-electron-concentration (VEC) effect of the transition metals (the two rationales are independent). As a result, **three Ti-based three-metal rocksalt carbides (hec3-MoTiZr/CrTiV/CrTiZr, (100)) passed the DFT dual objective**, confirming that H poisoning is the dominant failure mode and that low VEC is the physical lever that jointly tunes H repulsion and CO evolution. Full ionic relaxation of the representative candidate hec3-MoTiZr *confirms the selectivity* ($\Delta G_{\text{H}^*} = +0.57$ eV, actually strengthened; $\Delta G_{\text{CO}^*} = +0.18$ eV) while *correcting the limiting potential to $U_L = -0.40$ V*, showing that the single-point calculation had underestimated the *COOH barrier. Finally, re-aiming the generator at low VEC caused *the generative model itself* to yield H-repelling, CO-evolving winners, demonstrating that discovery can come from conditional generation rather than a post-processing screen.

To test these predictions, an experimental validation program consisting of synthesis, electrochemical CO₂RR, online gas analysis, ¹H NMR, and in-situ FT-IR was designed (Chapter 5). The broader contribution is to show, reusably, that economics, *synthesizability*, and *HER selectivity* can all be taken as *generative* design goals in catalyst discovery, that synthetic reality (acid leaching) defines the elemental palette, and that physics (hydrogen poisoning) defines the conditioning lever (VEC).

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